# Intangible Liabilities*

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**Abstract**

When liabilities are deemed improbable or cannot be reliably estimated by management, they are not recorded on the balance sheet. Instead, they are disclosed qualitatively in the company's filings. Examples include obligations related to pending or future lawsuits, product liability, environmental matters, false advertising, or patent and copyright infringements. We refer to these obligations as intangible liabilities (IL). We construct a firm-level, text-based measure of IL. IL is positively related to firm size, volatility, and share turnover, and is negatively correlated with accounting performance, abnormal returns, and Tobin's Q. IL also predictably varies across industries. Moreover, IL predicts future lawsuits against firms and the future deterioration of their reputations. Companies with higher IL trade at lower valuation ratios and have significantly higher future crash risks. A portfolio that is long on high IL and short on low IL yields an annual abnormal return of 3% after accounting for common factors. Overall, the results suggest that intangible liabilities are a significant determinant of firm value and stock returns.

Keywords: Intangible liabilities, market efficiency, litigation, firm reputation

JEL: G11, G12, G14, G32

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*More than 100,000 U.S. military veterans have filed lawsuits against Minnesota-based 3M over hearing damage linked to what they claim are defects in the company's military-grade earplugs.[…]The company faces billions of dollars in potential costs from legal settlements or trials.[…]In financial disclosures, 3M has said it hasn't set aside any money for settlements related to the earplug case because it says the liability isn't probable and that it can't reasonably estimate potential costs. 3M also said its reputation and cash flow could be hurt by product liability lawsuits.*

The Wall Street Journal (June 26, 2022)

*Bloomberg Intelligence estimates that the [3M] stock's valuation relative to its manufacturing peers implies a discount of about $33 billion tied to the legal battles. […] JPMorgan's [Stephen] Tusa this month increased what he called a conservative estimate of 3M's liabilities to a range of $2 billion to $185 billion [..]For its part, 3M hasn't reserved any funds to cover liabilities from the trials because "any such liability is not probable and reasonably estimable at this time."*

Bloomberg (February 13, 2022)

## 1. Introduction

Intangible assets are assets that are economically important (e.g., R&D, human capital, or brand value) but accounting rules do not permit them to be capitalized on the balance sheet. A growing literature focuses on measuring these assets and studies their relationship with different outcome variables. Since intangible assets are inherently difficult to measure, scholars have proposed and used different approaches, methodologies, and assumptions to obtain an estimate of different types of firm-level intangible capital (Edmans 2011; Eisfeldt and Papanikolaou 2013; Belo et al. 2022; Boustanifar and Kang 2022; Ewens, Peters, and Wang 2025; Iqbal et al. 2025; Boustanifar and Kang 2024). Overall, this literature suggests that intangible assets have important implications for firm value and stock returns.

In contrast, this paper studies intangible *liabilities*, which we define as economically significant liabilities that are not reported on the balance sheet.[1] As illustrated by the quoted text about 3M at the beginning of this article, a company could potentially face tens of billions of dollars in liabilities, even though its balance sheet shows no such obligation. According to accounting standards, while all material information related to potential liabilities must be disclosed in the text

[1] Merriam-Webster provides three definitions for the word *tangible*: "(a) capable of being perceived, especially by the sense of touch, (b) capable of being precisely identified or realized by the mind, and (c) capable of being appraised at an actual or approximate value (e.g., tangible assets)." Our definition of (in)tangible assets and liabilities align with the third definition, not the first one. Importantly, accountants also determine whether to include assets on the balance sheet not based on their physical presence but primarily on their valuation. For example, human capital (e.g., engineers), while physically present, is not included on the balance sheet. In contrast, cryptocurrencies owned by a firm, though not touchable, are included because their value can be determined. Moreover, acquired intangible assets with a readily available valuation are reported in the balance sheet.



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of company reports, only liabilities that are probable *and* estimable are recorded on the balance sheet. As a result, a significant portion of potential liabilities are disclosed in the text but unrecorded in the balance sheet. These liabilities can arise from various sources, including pending or future lawsuits, contract or patent infringements, environmental liabilities, or false advertising claims.

Intangible liabilities may represent a significant claim on firm assets, one that is senior to equity holders' claims, and could therefore impose a substantial negative effect on equity returns. For instance, as reported by Bloomberg, 3M's shares declined by 60% from 2018 to June 2023, wiping out approximately $100 billion in market value, primarily due to liability concerns.[2] Across the US economy, firms exposed to class-action lawsuits saw their aggregate market capitalization decline by an average of $989 billion during 1997 and 2021, and by $2.43 trillion in 2022.[3] Moreover, the disclosure of lawsuits leads to reductions in the value not only of the defendants' stocks (Bhagat, Bizjak, and Coles 1998) but also their peers' (Gande and Lewis 2009). This implies that the effect of intangible liabilities likely co-varies across firms. Indeed, intangible liabilities frequently stem from issues that affect multiple firms simultaneously, such as direct or indirect involvement in producing, selling, or using products and ingredients alleged to have negative effects on people or the environment. Therefore, intangible liabilities pose a significant risk to equity holders, and investors should, at least theoretically, demand higher expected returns to hold companies with greater intangible liabilities. However, despite their significant economic impact, to the best of our knowledge, no prior study has focused on measuring intangible liabilities and examining their effects on firm value and stock returns.

This paper proposes a text-based measure of intangible liabilities. To achieve this, we develop a dictionary of 569 related unigrams and bigrams, referred to as the IL dictionary, which are used in companies' annual reports. Examples include terms like *infringe*, *filed cases*, *filed suit*, and *environmental liability*. Our measure is based on the simple premise that companies with more such potential liabilities will exhibit a higher frequency of words and phrases related to intangible liabilities. It is important to note that companies are required by law to disclose their exposure to potential liabilities and hence do not have an incentive to hide them (unless when attempting to commit fraud). In fact, to avoid potential scrutiny from plaintiffs' lawyers, companies tend to

[2] "3M Heads to Trial in 'Existential' $143 Billion PFAS Litigation," Bloomberg, June 2, 2023.
[3] Cornerstone Research, Securities Class Action Filings, 2022 Year in Review.



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disclose potential liabilities as much as possible, though often vaguely and without quantitative estimates (Clor-proell and Maines 2014).

We use web crawling to download the universe of U.S.-listed companies' annual reports from the SEC EDGAR platform for the fiscal years between 1996, the first year in which electronic filings were mandatory, and 2020. After matching these firms with the Compustat for accounting variables and CRSP for market variables, we derive a sample of 112,716 firm-year observations. We then construct our measure of intangible liabilities (IL) for each of these firm-year observations, based on the frequency of words from our IL dictionary appearing in each filing.

We demonstrate that IL produces a reasonable ranking of industries based on their intangible liabilities. For instance, some of the usual suspects such as Tobacco, Insurance, Healthcare, and Pharmaceutical Products rank at the top of the list. At the lower end, industries such as Textiles, Printing & Publishing, and Steel appear.

While the average industry-level IL is informative, it conceals significant within-industry heterogeneity across firms. In nearly every industry, we find companies with either low or high intangible liabilities. For example, while Agriculture and Business Supplies are generally low IL industries, companies like Cultiva and 3M, within these sectors, rank among the highest IL companies. These results indicate that while industry plays an important role in explaining intangible liabilities, a significant variation in IL remains unexplained by industry-specific characteristics. Indeed, our regressions show that up to 13% of variation in IL can be attributed to industry factors.

Additional analyses provide further validation for the IL measure. For instance, we demonstrate that there is a reasonable correlation between IL and various firm characteristics. Firm size, volatility, and turnover are positively correlated with IL, whereas performance and Tobin's Q show a negative correlation. Furthermore, companies with higher IL trade at lower valuation ratios (i.e., P/E or M/B), suggesting that investors consider these liabilities when valuing firms. In addition, companies with higher IL are more likely to face class action lawsuits in the future. Consistently, the IL measure is significantly related to future deterioration of firms' reputations, as indicated by negative media coverage.

In the next step, we analyze the stock returns of companies with varying levels of IL. Our findings indicate that investors demand higher risk premia to invest in firms with greater exposure



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to IL. We demonstrate this difference in risk premia by constructing an IL portfolio, which is long on firms with high IL and short on those with low IL. This long-short IL portfolio generates a significant annual average return of 4.35% (or 3% after adjusting for industry returns). This dispersion in expected returns due to IL heterogeneity is not explained by the CAPM, the Fama-French 3-factor (FF3) model, the 5-factor of Fama and French (2018) plus momentum (FF6), or the augmented q-model of Hou et al. (2020) (q5). Depending on the model, we obtain significant annual alphas for the IL portfolio, ranging from 2.91% (*t*-statistic, 2.07) in the q5 to 5.47% (*t*-statistic, 3.24) in the FF3 model. Moreover, controlling for exposure to "mispricing" factors, as in Daniel, Hirshleifer, and Sun (2019) or Stambaugh and Yuan (2016), results in even larger abnormal returns, ranging from 5% to 6%. These results are robust to adjusting for industry returns. Finally, using Fama-MacBeth regressions and controlling for a host of well-known factors and characteristics including different measures of intangible assets, we find consistent results: IL serves as a robust signal for predicting future stock returns.

Investors appear to demand a higher risk premium to invest in companies with greater exposure to IL, as intangible liabilities could suddenly materialize (e.g., if a firm loses a major lawsuit), leading to a significant drop in value. In other words, companies with substantial intangible liabilities are expected to face a higher risk of stock price crashes. We test this hypothesis and find that indeed companies with higher exposure to IL do, in fact, face significantly higher future stock price crash risk. This result remains robust across various measures of crash risk and holds across multiple specifications. We also test the mispricing hypothesis but find no evidence to support it.

Our paper relates to the literature that attempts to measure intangible *assets* and their stock market valuation. Due to the inherent difficulty of measuring intangible assets, prior studies have relied on various assumptions and methods to propose measures that, while imperfect, capture certain aspects of intangible assets (Edmans 2011; Eisfeldt and Papanikolaou 2013; Belo et al. 2022; Boustanifar and Kang 2022; Ewens, Peters, and Wang 2025; Iqbal et al. 2025; Boustanifar and Kang 2024; Gulen et al. 2025). In this paper, we take the first step in measuring firm-level intangible *liabilities* and examine their impact on firm value and stock returns.

Our paper also relates to the body of work by accounting scholars on measuring litigation risk, which refers to the risk of a firm facing future lawsuits, using various methods (Francis, Philbrick,



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and Schipper 1994; Kim and Skinner 2012; Lee et al. 2020; Kartapanis and Yust 2024). The likelihood of a firm facing future lawsuits is an important, but not the only, source of IL. Specifically, IL can arise from other sources including, but not limited to, pending lawsuits, product recalls, loan guarantees, environmental liabilities, and pending legislation.[4]

We also contribute to the growing literature that employs textual analysis of annual filings (or conference calls) to construct hard-to-measure firm-level characteristics. A few examples include measuring competition (Li, Lundholm, and Minnis 2013), accruals (Frankel, Jennings, and Lee 2016), corporate culture (Li et al. 2020), brand value (Boustanifar and Kang 2024), financial constraints (Hoberg and Maksimovic 2014; Buehlmaier and Whited 2018), firm's involvement in "sinful" activities (Boustanifar and Schwarz 2025), and cybersecurity risk (Florackis et al. 2022). We contribute to this literature by proposing a simple and transparent method to measure intangible liabilities.

The remainder of the paper is organized as follows. Section 2 describes the data, measurement, and descriptive statistics. Section 3 examines the variation in intangible liabilities across and within industries. Section 4 explores which firm characteristics are correlated with intangible liabilities. Section 5 analyzes the relationship between intangible liabilities, valuation ratios, and future lawsuits, while Section 6 explores the connection between intangible liabilities and risk premia using factor regressions and the Fama-MacBeth setup. Section 7 studies the relationship between intangible liabilities and future stock price crash risks. Section 8 analyses different sections of the 10k filings, and Section 9 investigates the mispricing hypothesis. Section 10 identifies the most important words in terms of predictive power. Section 11 provides the conclusion of the study.

## 2. Data, measurement, and descriptive statistics

### *2.1. Data sources and sample construction*

We use two different samples in this paper. Our main sample consists of all US firms for which we could download 10-K annual reports from the SEC EDGAR platform between fiscal years 1996 and 2020. After matching these firms with Compustat (for accounting variables) and CRSP (for

[4] We show that our conclusions hold when controlling for litigation risk measures proposed in the literature.



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market variables) databases, we derive a sample of 112,716 firm-year observations. We refer to this sample as the core sample. We construct the text-based measure of intangible liabilities (IL) for each of these firm-year observations.

We also gather data on class action lawsuits and merge it with our dataset to study whether IL predicts future lawsuits. To do so, we manually collect data provided by the Securities Class Action Clearinghouse (SCAC) maintained by Stanford University (securities.stanford.edu). This is the most comprehensive database used in the literature for this purpose (Arena and Julio 2015; Kim and Skinner 2012). We found 5,911 class action lawsuits during our sample period. While many lawsuits are filed against private firms not included in our analysis, about half of them (2,821) concern companies in our sample. For comparison, Kim and Skinner (2012) find 720 lawsuit filings matched with CRSP and Compustat data in their sample that ends in 2008.

Our second sample, which we refer to as the RepRisk sample, is used to study whether IL predicts firm-level reputation risk/loss. This analysis relies on data provided by RepRisk, a database that gathers information on the negative exposure of firms with respect to (sustainability) activities by screening more than 100,000 sources (e.g., print, online, and social media; regulators; think tanks; newsletters; blogs). RepRisk considers 28 topics to estimate overall firm-level reputation risk including, for example, violation of laws, regulations, and standards; occupational health and safety; anti-competitive practices; fraud; health or environmental issues caused by products or services; poor employment conditions; child labor; pollution; and tax evasion.

RepRisk uses the negative coverage and a proprietary algorithm (considering the type of media coverage, the severity of the incidents and their timeliness, among others) to compute the RepRisk index (or RRI), which captures the extent to which a firm is exposed to reputational risk. The RRI ranges between 0 and 1 (multiplied by 100, in our analysis), in which higher values indicate higher reputation risk. The RRI indicator is available on a monthly basis, but in our analyses, we use the RRI of the fiscal year's last month. These data have a shorter historical coverage than our main sample. While RepRisk data starts from 2008, the coverage is low in early years. We purchased data from the RepRisk provider for the period January 2012 through December 2021 for 1,669 unique American-listed companies. Merging with the core sample and requiring the availability of RRI and market capitalization information leads to a sample of 11,329 firm-year observations.



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*2.2. Text-based firm-level intangible liabilities*

We use web crawling to download all annual reports from the SEC EDGAR platform for fiscal years 1996 to 2020. We consider regular annual reports (10-K), annual reports with checked item 405 (10-K405), annual reports for smaller businesses (10-KSB), and transitional annual reports (10-KT) for companies that change their fiscal year. When companies file multiple annual reports in the same year, we keep the main filings and only use amendments (e.g., 10-K/A) when there is no main filing. When there are multiple main filings (i.e., a 10-K and a 10-KT or a 10-KSB), we prioritize the regular annual reports (10-K). We then extract the text of the filings and after standard pre-processing such as removing stop-words (such as "a," "to," "and," etc.) and exhibits, we keep filings with at least 2,000 words.

Our text-based measure of intangible liabilities relies on the premise that companies are required by the SEC to disclose (in different sections of their annual reports) possible circumstances or situations that make investing in a company's security risky. An important part of these risks is contingent liabilities, that is, possible obligations/losses that may occur depending on the outcome of an uncertain future event. It is important to note that these liabilities are not recorded on the balance sheet. Provisions reported on the balance sheet are liabilities that are likely to occur, *and* their amount can be measured reliably. If at least one of these conditions is not met, then the liability will not enter the balance sheet and instead must be disclosed (in text) in the annual report. Some common examples of intangible liabilities include pending or potential lawsuits; product warranties; infringement of contracts, copyrights, or patents; environmental liabilities (e.g., due to hazardous wastes and pollution); and pending investigations. For instance, firms face or could face lawsuits from: (i) competitors for infringements of patents or copyrights; (ii) communities regarding environmental matters, pollution, and waste management; (iii) regulators for violating laws; and (iv) (a group of) individuals for being injured (financially or personally) because of using the company's products or services.[5]

[5] Our definition of intangible liabilities includes but is not limited to contingent liabilities disclosed by firms. For example, if the management believes that a contingency is not reasonably possible, it will not be disclosed as a contingent liability. It could also be that a firm's business is inherently exposed to a particular type of liability (e.g., personal injury or environmental liability) but has not yet been the subject of any related lawsuit. Such an exposure to potential liabilities is disclosed in other sections such as risk factors. This is why it is important to analyze the whole filings rather than a particular section of them.



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To construct a text-based measure of firm-level intangible liabilities, we propose a dictionary of unigrams (single words) and bi-grams (sequence of two words) that have appeared in annual reports and are related to intangible liabilities. Evaluating alternative methods to measure the tone in annual reports, Henry and Leone (2016) conclude that "although more complex techniques are potentially advantageous in certain contexts, equal-weighted, domain-specific, word frequency tone measures are generally as powerful in the context of financial disclosure and capital markets." However, they highlight that for a dictionary-based approach to perform well, one needs to exclude potentially problematic words, those that are likely to appear in other contexts. Consequently, our aim is to create a dictionary, using domain-specific knowledge, that includes words and phrases that are closely related to intangible liabilities.

We start with an existing and somewhat relevant dictionary. While the influential work of Loughran and McDonald (2011), hereafter LM, was focused on developing a list of negative words in the context of company reports (and showing that it outperforms standard negative lists used by researchers), they also proposed several other word lists such as uncertain, strong, and weak modal, constraining, and litigious. The LM litigious dictionary includes 904 words (unigrams), which serves as our starting point. We carefully analyzed every single word in the LM dictionary and removed all words (407 out of 904 words) that are not directly linked with intangible liabilities and/or those that do not have a negative connotation. Examples of words we removed from the LM dictionary because they are irrelevant in our context are *abovementioned*, *acquirors*, *admission*, *aforementioned*, *amend*, *codification*, *contract*, *contractholder*, *designator*, *extracorporeal*, *hereby*, *hereafter*, *interrogate*, *moreover*, *shall*, *whatsoever*.

While it is obvious in some cases to exclude a word (see some examples above), a careful analysis of the context in which a word is typically used in the annual filings was necessary in deciding to exclude many other words. Here we provide a few examples: *interrogate/interrogator* is often used in biotech companies such as "our platform allows us to interrogate B cells…" and "The interrogator is the optics/electronics…" *Extracorporeal* is a medical procedure performed outside the body and hence it is often used in that context. *Designator* is often used in airlines because designators are abbreviated codes that are internationally recognized and used to identify airlines. We also removed some words in the LM dictionary that never showed up in any filing, such as *provisos, predeceasing, libels, interpositions, incarcerates, executrixes, codicils*.



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More importantly, we added 72 new words and phrases that are commonly used by companies and that contain information about firms' exposure to intangible liabilities. Examples of such words include *infringe, civil payment, counter claim, counterfeit, environmental liability, fraudulent, hazardous wastes, liable, misrepresentation, product liability, unfair competition, false advertising, filed cases, filed suit,* and *professional liability.* We note that unlike the LM dictionary, our dictionary includes bigrams, which is an important feature in correctly identifying information about intangible liabilities. For example, the LM dictionary correctly excludes *liability* as it does not have a negative connotation in the context of company filings, but *environmental liability* or *product liability* are clearly informative about companies' intangible liabilities, and these are bigrams included in our dictionary.

Overall, our dictionary includes 569 words/phrases and is quite different from and more suitable for our purpose than the LM litigious dictionary as discussed above. Having said that, it is important to note that the focus of LM and our paper are completely different; LM was focused on proposing a list of negative words that measures the tone of company reports better than the standard dictionaries.[6] We, on the other hand, focus on proposing a method to measure intangible liabilities, validating the measure, and study its relationship with firm value and future stock returns, none of which is studied in LM. Appendix Table 1 provides the full list of words and phrases in our IL dictionary.

Using the IL dictionary, we count the frequency of words in our dictionary for each firm-year observation, and our main measure of intangible liabilities (IL) is the log of frequency counts. In computing the total frequency of IL words, we use the equal-weighted method and do not weight different words differently (Henry and Leone 2016). Moreover, while our results are robust to using the frequency of IL words as a percentage of total words, we use the log frequency as our main measure. The former is common and reasonable in cases such as sentiment analysis in which every word in the filing could be either categorized as positive, negative, or neutral. In our case, however, we are looking for a particular type of information in the filings, and dividing the counts of those

[6] Almost all the analyses in LM are based on the negative words and they study returns in a 4-day window around the filing date. Table VI of LM is the only one that studies excess returns around the filing date for other word lists and interestingly they do not find any relation between filings with more litigious words and returns around the filing date. In unreported regressions, we do not find any robust relation between measures constructed using the LM litigious dictionary and future lawsuits, reputation risk, or stock returns.



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words by the total word counts would generate a metric whose variation is mostly driven by differences in word counts rather than differences in IL-related words. Of course, it is important to control for the length of filings in regressions, and we do so in all our analyses. Whenever it is not possible to control for the length of the filings (e.g., in the time-series portfolio analysis), we use the frequency of IL words as a percentage of total words.

### *2.3. Descriptive statistics*

Table 1 reports descriptive statistics of the main variables used in the paper. Our main variable of interest, *IL*, has an average (median) of 4.64 (4.79), with a minimum of 0 and maximum of 8.58. To better understand the distribution of this metric, Figure 1 plots the histogram of this variable. As also shown in Table 1, on average, each filing includes 39 unique words from the IL dictionary, while the maximum number of unique words in a filing in our sample is 149. About 2% of firms were involved in a class action lawsuit during our sample period.

To provide more detailed descriptive statistics for the core sample as well as corresponding statistics for the RepRisk sample, Table 2 reports similar statistics as in Table 1 for each quintile of the IL measure (Panel A) as well as a corresponding table for the variables used in the RepRisk analysis (Panel B). It is important to note that these are simply descriptive statistics and should not be used to draw causal inferences. With that in mind, we observe that companies with higher IL are larger, have lower ROA and tangibility, and have higher leverage, volatility, and turnover. We do not see a monotonic relationship between valuation ratios (PE and MB) and IL, and if anything, there seems to be a positive association between them in the raw data. As will be shown later, we do observe a strong (negative) correlation between them in regressions when we control for firm and industry characteristics.

Importantly, Table 2 also shows that companies with higher IL are much more likely to be the subject of class action lawsuits (0.32% versus 5.66% for firms in the first versus fifth quintiles). We observe somewhat similar patterns related to the abovementioned variables in the RepRisk sample in Panel B. Interestingly, again we see a significant increase in reputation risk or RRI (from 8.45 to 14.34) as we go from the first to the fifth quintile of IL.



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## 3. Intangible liabilities across and within industries

A natural way to check the validity of the IL measure is to compare the metric for firms across different industries. An important factor explaining intangible liabilities is the industry in which the firm operates. For instance, it is common knowledge that firms in some industries (e.g., tobacco, pharmaceuticals) inherently face more intangible liabilities than those in some other industries (e.g., business supplies, agriculture). These industries produce products that have or are likely to have adverse effects on consumers, leading to lawsuits. Their practices are also typically subject to more laws and regulations and hence a higher likelihood of violating laws and regulations.

Figure 2 plots the average frequency of IL-related words per each Fama-French 49-industry classification. Indeed, overall, the results are consistent with what one would expect. For example, we find that Tobacco Products, Insurance, Healthcare, and Pharmaceutical Products are ranked within the top four industries in terms of intangible liabilities.[7] Automobiles, Restaurants and Hotels, and Food Products are in the middle range of the distribution, whereas Textiles, Printing and Publishing, and Apparel appear to have lowest intangible liabilities, on average. Perhaps surprisingly, Beer & Liquor also shows up as an industry with very low intangible liabilities even though these companies are considered within sin stocks (Boustanifar and Schwarz 2025). We double-checked the company filings and indeed an *average* firm in this sector (e.g., the Boston Beer Company) carries a relatively low level of intangible liabilities.

We then proceed to study within-industry distribution of intangible liabilities. We find that even in industries with an overall low intangible liability, there are firms with very large risk. Figure 3 plots five top companies in Business Supplies and Agriculture in terms of intangible liabilities. As shown, while companies in Business Supplies have, on average, quite low risks, one company within the same industry, namely 3M, appears to have significant intangible liabilities. Indeed, 3M has been the subject of large controversies around personal injury (as well as arguably adverse environmental and health effects) from the perfluorinated compounds such as PFOA and PFOS

[7] It is important to note that Utilities also rank among top IL industries ($6^{th}$). We manually checked, and indeed some companies in this sector have high litigation risk. PG&E, for example, filed for bankruptcy twice in our sample period (in 2001 and 2019) because of their intangible liabilities such as facing potential liabilities of over $30 billion from multiple wildfires in the years 2015–2018. The Cornerstone Research on Securities Litigation (Year 2022 in Review) also shows that Utilities and Financials historically have been among the top sectors in terms of lawsuits, after Healthcare. However, we also show later that there are large within-industry variations in IL.



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used in its products. As per the company's filing, there are thousands of individual and class action lawsuits filed against the company related to different products and practices.

A similar case is presented in Figure 3 for the agriculture industry, where Corteva (the agricultural unit of DowDuPont prior to being spun off as an independent company) shows up with very high intangible liabilities. The company indeed has a huge number of lawsuits and potential liabilities because of them. Among entities suing the company, according to its 2021 report, are the Canadian Competition Bureau, The Chemical Safety Board, and the U.S. Environmental Protection Agency; the U.S. Federal Trade Commission also is investigating the company for unfair trading. The company also lists numerous pending personal injury and remediation lawsuits related to exposure to, or contamination by, the company's well-known insecticide called chlorpyrifos, and potentially significant contingent liabilities related to the company's environmental impacts (such as hazardous waste management, chemical contamination harming natural resources, and contamination of public water systems).

The abovementioned cases suggest that there might be substantial within-industry variation in intangible liabilities across firms. To investigate within-industry differences in intangible liabilities, Figure 4 plots the distribution of IL for firms within each industry. The main message of this plot is that due to large within-industry variation in intangible liabilities, it is extremely difficult to differentiate high-IL versus low-IL industries (by only looking at firm-level metrics). We manually checked companies in several industries and indeed companies in the same industry, and sometimes seemingly similar companies, have quite different intangible liabilities (e.g., IBM has very low IL, whereas HP has very high IL).

Overall, the results reported above suggest that while part of intangible liabilities is industry-specific, a significant fraction is firm-specific, something we will formally investigate in the next section.

## 4. Firm characteristics and intangible liabilities

In this section, we investigate whether (and which) firm and industry characteristics are correlated with our measure of intangible liabilities. In addition to the usual suspects (e.g., size, performance, asset tangibility), we include variables that have been argued in the prior literature



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(Kim and Skinner 2012) to be related to litigation risk including abnormal returns, return volatility, and share turnover.

To isolate the macro shocks or secular increases in disclosure details over time, all our regressions include year fixed effects. Moreover, we add a set of industry effects (Fama-French 49 classifications) to control for differences in IL across industries. In additional specifications, we replace industry effects with firm fixed effects to study within-firm variation in intangible liabilities. Before we report the results, it is important to highlight that our regression analysis in this section only aims at capturing correlations and should not be interpreted as causal or predictive relationships.

Table 3 reports the results of regressions investigating which firm characteristics correlate with our measure of intangible liabilities, IL. The first three columns include year and industry fixed effects and are different only in terms of the set of control variables used. Standard errors clustered at the industry level are reported in parentheses. The results in column (1) show that industry, year, and size together explain about half of the variation in intangible liabilities. In unreported regressions, we find that industry effects alone account for 13% of the total variation in IL. Overall, these results show that industry, size, and macroeconomic variables are important in explaining variations in IL, as one would expect.

However, the regressions also show that about half of the variation in IL cannot be explained by macro variables, industry, and size. We add additional firm-level variables in columns 2 and 3. Firms with high IL tend to have lower accounting performance (ROA), leverage, abnormal returns, and Tobin's Q, while having higher volatility and turnover. These results support the approach used by Kim and Skinner (2012), who use abnormal return, volatility, and turnover to construct firm-level litigation risk. We find similar relationships even when we study within-firm changes in intangible liabilities in column 4 (except for size and abnormal returns whose coefficients become statistically insignificant).

Overall, the results reported in Table 3 show that while industry characteristics, macroeconomic variables (captured by year effects), as well as several other firm-level variables have power in explaining variations in IL, a significant part remains unexplained by these factors.



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## 5. Intangible liabilities, current valuation, and future lawsuits/reputation

### *5.1. Intangible liabilities and valuation ratios*

In this section, we study whether and how IL and current valuations are related. We expect that, *ceteris paribus*, companies with higher IL are traded at lower valuations. To investigate this relationship, we run regressions with measures of valuation ratios (price-to-earnings or market-to-book) as the dependent variables and IL as the main independent variable. We also control for important firm-level variables that are expected to be related to firm valuation such as size, leverage, volatility, profitability, and tangibility. To control for differences across industries and macro conditions affecting valuation ratios, we also include industry and year effects in regressions.

Table 4 reports the estimated effect of IL on the PE ratio (columns 1-2) and the MB ratio (columns 3-4). The difference between the two specifications for each of the dependent variables is the choice of fixed effects. In columns 2 and 4 (compared to columns 1 and 3), we replace industry effects with firm fixed effects. The results show that IL and valuation ratios are negatively correlated; the higher the IL, the lower the valuation ratios. The estimated coefficients suggest that a one standard deviation increase in the frequency of IL related words is associated with about a 4-5% decrease in valuation ratios. These results provide both support and validation for our proposed measure and show that the market does consider intangible liabilities in valuing companies.

### *5.2. Intangible liabilities and future lawsuits*

In this section, we study whether IL has predictive power in explaining future lawsuits. To that end, we run logit and linear probability models in which the dependent variable is a lawsuit dummy that takes a value of 1 if a class action lawsuit is filed against the firm within the next year and 0 otherwise. In these regressions, we also add several variables that have been proposed to be related to the likelihood of being sued, such as abnormal returns, volatility, and turnover (Kim and Skinner 2012).

Table 5 reports the relation between IL and future lawsuits, using both logit (Panel A) and linear probability (Panel B) models. All regressions include year fixed effects. In addition, columns



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1-4 include industry fixed effects, whereas columns 5-8 replace them with firm fixed effects. Otherwise, differences across specifications are related to the set of control variables used.

As shown, regardless of the regression model or specification used, we find a strong relationship between IL and the probability of future lawsuits. Our most conservative estimate of 1.47 (column 3, Panel B of Table 5) implies that a one standard deviation increase in IL is associated with 1.64 percentage point higher probability of a future lawsuit. Given that the unconditional probability of receiving a lawsuit is 2.2% in the sample, the magnitude of the relationship is quite large. In addition, the robust and significant results of columns 5-8, including firm fixed effects, show that an increase in IL within a firm also predicts a higher likelihood of the same firm being sued in the future. Overall, the results shown in Table 5 provide further validation for the IL measure. These results also complement the literature on litigation risk by showing that in addition to variables used in Kim and Skinner (2012), the contents of annual filings and their changes contain important forward-looking information related to the likelihood of being sued in the future.[8]

### *5.3. Intangible liabilities and future reputational risk*

In this section, we study the relationship between IL and the future reputation risk faced by the company. We measure an individual firm's exposure to reputation risk using data provided by RepRisk, which uses the firm's negative exposure in local, national, and international media to capture its reputation risk. To investigate how IL relates to future reputation risk, we run regressions with the reputation risk index, RRI, as the dependent variable. RRI ranges from 0 to 1 (or 100%), with larger numbers being associated with higher reputational harm. Importantly, we investigate the relation between current measures of IL and future RRI, to gain insights on whether IL is able to predict reputational harm for companies in the future, inflicted by the media. Specifically, we use eight specifications including different control variables, and year and industry (or firm) fixed effects, similar to the analyses reported in Table 5.

[8] In supplementary analysis, reported in Appendix Table 3, we demonstrate that our results remain robust after controlling for a measure of litigation risk, following the approach of Lee et al. (2020). We note that this adjustment leads to a significantly smaller sample size due to the limited availability of litigation risk data.



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Table 6 provides results of the relationship between IL and future reputation losses. We find a strong positive relationship between the two, which is robust across different models and specifications. The most conservative estimate from within-industry results, i.e., the coefficient of 1.88 in column 3, implies that a one standard deviation increase in IL is associated with an RRI increase in the following year (or, future reputation loss) of 2.2. As the unconditional mean for RRI is 10.4 in our sample, this represents a 20% increase in reputation loss of an average company. We find similar results in specifications using firm fixed effects (columns 5-8) showing that changes in IL within firms also predict changes in the firm's reputational risk.[9]

Overall, the results reported so far show that firms with a higher text-based measure of IL are traded at lower valuations, have a higher likelihood of being sued, and are more likely to face reputational harm.

## 6. Intangible liabilities and risk premia

### *6.1. Time-series factor regressions*

In this section, we examine whether cross-sectional differences in IL are associated with differences in risk premia. To address this, we keep stocks with a December fiscal year-end and sort them at the end of June of each (year t) based on IL, constructed from financial statements for the previous fiscal year (year t-1). This approach follows the standard in the literature to ensure that the information used to construct the IL measure is publicly available at the time of portfolio formation. As noted earlier, since controlling for the total number of words in each filing is not feasible in a time-series portfolio analysis, the IL measure in this section is based on the frequency of words from the IL dictionary as a percentage of total words in each filing. We then create five portfolios, ranked from 1 to 5, where portfolio 1 (5) contains the stocks in the bottom (top) 20$^{th}$ percentile of IL. Monthly returns and market capitalization data from CRSP are used to compute value-weighted portfolio returns.

[9] In supplementary analysis reported in Appendix Table 4, we also show that companies with higher IL have higher cost of debt and audit fees.



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Panel A in Table 7 shows that there is a significant difference (4.35% with *t-statistic* 2.49) in the average annual returns of portfolios with high versus low IL. The second row of Panel A shows that the significant difference in these returns remains even after adjusting for industry returns.

To investigate whether the difference between portfolios with high and low IL is driven by well-known risk factors, we form a portfolio that is long stocks with high IL (top quintile) and short stocks with low IL (bottom quintile), which we refer to as the IL portfolio. We then estimate the alpha (or excess returns) of the IL portfolio using different factor models. Specifically, we use the CAPM, Fama-French 3-factor (FF3), the 5-factor of Fama and French (2018) plus momentum (FF6), and the q5 model of Hou et al. (2020). In addition, to investigate whether the excess returns are due to exposure to mispricing factors, we estimate abnormal returns using the "mispricing" factors of Stambaugh and Yuan (2016) [SY] and the long- and short-horizon behavioral "mispricing" factors of Daniel, Hirshleifer, and Sun (2019) [DHS].[10] Overall, we report abnormal returns of the IL portfolio using 6 factor models.

The annualized excess returns of the IL portfolio using various factor models are reported in the first row of Panel B of Table 7. We find economically sizable and statistically significant excess returns across all models. For example, the FF6 alpha is estimated to be 3.45% (*t*-statistic, 2.72). We find larger alphas (5-6%) when we use mispricing models of DHS and SY. The most conservative estimate comes from the q5 model with an annualized alpha of 2.91% (*t*-statistic, 2.07). In unreported regressions, we run an alternative model including common factors of market, size, value, momentum plus the betting against beta factor of Frazzini and Pedersen (2014) and quality minus junk factor of Asness, Frazzini, and Pedersen (2019). The estimated alpha in this 6-factor model is 6.47% (*t*-statistic, 4.44).

We conduct further analyses to better understand the IL abnormal returns. First, as shown in earlier sections, IL varies across industries. One might wonder how much of the premium is due to affiliation to a particular industry. To address this, we re-estimate alphas using the industry-adjusted returns. The results reported in the second row of Panel B of Table 7 show positive and significant industry-adjusted alphas across all models. For instance, FF6 industry-adjusted alpha is

[10] We note that tables related to this section report the *t*-statistic using Newey standard errors (not the standard errors themselves) in parentheses below the coefficients, as it is more common in the asset pricing literature to report *t*-statistic rather than standard errors.



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2.99% (*t*-statistic, 2.43). In addition, to better understand time series of abnormal returns, Figure 5 plots the cumulative excess return (alpha plus residuals) of the IL portfolio over time using the main four factor models discussed above. As shown, it appears that abnormal returns are present throughout the sample period. In untabulated regressions, we also estimate excess returns for different periods: pre-2000 (.com boom), 2000-2002 (.com bust), 2002-2008 (pre-fin-crisis boom), 2008-2009 (fin-crisis), and after 2009. The estimated alphas are positive in all periods except for the financial crisis, which is negative (but not statistically significant). Our subsample analysis, therefore, does not reveal any systematic differences in excess returns across years. In our final supplementary analysis, we find that the excess return of the long-short IL portfolio is driven both by the long and short leg, as shown in the Appendix Table 5.

Overall, these results show that, after controlling for common factors including the mispricing factors, the IL portfolio generates a significant and positive annualized alpha of at least 3%. This, in turn, suggests the existence of an IL premium, that is, stocks with higher IL have higher returns.

### *6.2. Fama-MacBeth regressions*

One main advantage of time-series factor regression analysis reported above is that it does not assume any particular functional form for the relation between IL and future returns. However, aggregating to the portfolio level loses information in the cross-section of stock returns and it does not enable us to control for more than a handful of factors. In other words, the results of factor regressions could simply be due to exposure of the portfolio to other known risk factors and characteristics (e.g., skewness, R&D expenditures, advertising expenditures, idiosyncratic volatility) that predict the cross-section of stock returns.

To address this issue, we turn to estimating monthly stock-level cross-sectional Fama-MacBeth regressions. We start with the universe of common stocks in CRSP (share codes of 10 and 11) that are listed in one of the main three main U.S. exchanges: NYSE, NASDAQ, and AMEX. The monthly stock returns data from CRSP are merged with accounting data from Compustat. This CRSP-Compustat merged dataset is then further merged with our IL data. Using this comprehensive dataset, we run cross-sectional regressions of stock returns in month t+1 on our measure of IL and other characteristics/return predictors, all of which are available at the end of



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month t. It is important to note that we have followed the standard practice in the literature by applying a six-month lag to accounting variables and IL, derived from annual filings, before merging with CRSP data. This ensures that the data is public and available to investors at the end of month t.

In Fama-MacBeth regressions, we control for beta, size, book-to-market ratio, momentum, profitability, asset growth, trade volume, idiosyncratic volatility, skewness, illiquidity, price, yield, advertising expenditures, R&D expenditures, leverage, accruals, repurchases, net equity issuance, as well as the length of companies' annual reports. All control variables are winsorized at the 1% level on both sides and we use standardized variables (demeaned and divided by their standard deviation) for easier interpretation of results. Variable definitions are reported in Appendix Table 2.

Table 8 reports the results of the Fama-MacBeth regressions, that is, the average coefficients from the monthly regressions together with their *t*-statistic based on Newey-West standard errors. Regardless of specification and the number of return predictors included, we find a robust predictive power for IL. Interestingly, the magnitude of the effect changes very little as we increase the number of controls in regressions. In column 6, including the complete set of characteristics, we find a time-series average of the cross-sectional coefficient of 0.16% (*t*-statistic, 4.17) for IL. This implies that a one standard deviation increase in IL leads to a monthly increase of 0.16% (or 2% a year) in stock returns.[11]

We conduct several additional analyses, which are reported in the Appendix. Specifically, in Appendix Table 6 and Appendix Table 7, we demonstrate that our results remain robust even after incorporating litigation risk measures introduced by Lee et al. (2020) and Kartapanis and Yust (2024), despite the significant reduction in sample size due to the limited availability of these measures. We find that the estimated coefficients on these litigation risk measures are not robust and are never positive and statistically significant when common control variables are included. Additionally, in Appendix Table 8, we show that controlling for various measures of intangible

[11] In Appendix Table 6 and Appendix Table 7, we demonstrate that our results remain robust when adding litigation risk measures introduced by Lee et al. (2020) and Kartapanis and Yust (2024), despite the significant reduction in sample size due to the limited availability of these measures. The estimated coefficients on these litigation risk measures are not robust, and they are never positive and statistically significant when common control variables are included.



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capital — such as the organization capital of Eisfeldt and Papanikolaou (2013) or the knowledge and organization capital of Peters and Taylor (2017) — does not affect the magnitude of our coefficient of interest but increases its statistical significance.

Overall, these results show that, controlling for a host of known characteristics, IL has predictive power in explaining the cross-section of stock returns.

## 7. Intangible liabilities and crash risk

The results of previous sections showed that companies with high IL have higher stock returns, even after controlling for known factors and characteristics that are related to stock returns. Presumably, higher returns for these companies are compensation for the risk of holding companies with high exposure to intangible liabilities. More specifically, intangible liabilities could become actual liabilities (e.g., when a firm loses in a class action lawsuit), in which case equity holders could face a significant loss. If that is the case, we expect to see a significantly larger likelihood of future stock price crash risk for companies with higher IL.

To investigate the relation between IL and *firm-specific* crash risk, we follow the prior literature (Chen, Hong, and Stein 2001; Kim, Li, and Zhang 2011b, 2011a; Hasan, Taylor, and Richardson 2022) and first estimate firm-specific returns. To do so, we estimate the following expanded market model regression:

$$r_{j,t} = \alpha_j + \beta_{1j} r_{m,t-2} + \beta_{2j} r_{m,t-1} + \beta_{3j} r_{m,t} + \beta_{4j} r_{m,t+1} + \beta_{5j} r_{m,t+2} + \varepsilon_{jt} \qquad (1)$$

where $r_{j,t}$ stands for return on stock $j$ in week $t$, and $r_{m,t}$ is the return on the CRSP value-weighted market index in week $t$. Similar to Kim, Li, and Zhang (2011b), we also include lead and lag terms for market return to allow for nonsynchronous trading. The firm-specific weekly return of firm $j$ in week $t$ (or $w_{j,t}$) is then computed by the natural logarithm of one plus the residual return from Equation (1), that is, $w_{j,t} = \ln(1 + \varepsilon_{jt})$.

Using these firm-specific returns, we follow prior literature mentioned above and use the negative conditional return skewness (NCSKEW) as our main measure of crash risk. For each firm



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in each fiscal year, this measure is computed by taking the negative of the third moment of $w_{j,t}$ over each year and dividing it by the standard deviation of $w_{j,t}$ raised to the third power.

We also use two additional measures of crash risk. Our second measure of crash risk (CRASH_EXTR) is the negative of the worst deviation of $w_{j,t}$ from the average $w_{j,t}$ scaled by the standard deviation of $w_{j,t}$ over the entire fiscal year (Andreou, Louca, and Petrou 2016). Following Kim, Li, and Zhang (2011b), our third measure of crash risk is an indicator variable that has a value of 1 if a firm experiences at least one crash week in a fiscal year and 0 otherwise. A crash week is a week during which $w_{j,t}$ is 3.2 standard deviations (the 0.1% cutoff of the normal distribution) below the average $w_{j,t}$ over the entire fiscal year.[12] On average, 18.48% of firms in our sample experienced at least one crash week in a year, which is in line with the 16.3% that Kim, Li, and Zhang (2011b) report for their sample.

We estimate regressions where the dependent variable is future firm-level crash risk (multiplied by 100), and our main righthand-side variable is IL. Following prior literature (Hasan, Taylor, and Richardson 2022), we include a set of control variables that are known to be related to crash risk, including lagged crash risk (to account for persistence in crash risk measures), de-trended trading volume (DTURN), the average $w_{j,t}$ over previous fiscal year (RET), standard deviation of $w_{j,t}$ over the fiscal year multiplied by 100 (SIGMA), the natural log of total assets (SIZE), the market-to-book ratio (MTB), total debt over assets (LEV), profitability (ROA), and accrual manipulation (ACCM), measured as the three-year moving sum of absolute discretionary accruals following the modified Jones model (Dechow, Sloan, and Sweeney 1995). In addition, we include both year and industry fixed effects to account for any macro shocks or differences in crash risk across industries. We cluster standard errors at the firm level.

Table 9 reports the estimated effects of (firm-level) IL on future firm-specific crash risk. We observe a positive and statistically significant effect of IL on future crash risk, which is robust across different specifications and using different measures of crash risk. The economic magnitude of this effect is also significant. For example, the coefficient of 0.98 in column 3 with the full set of control variables implies that a one standard deviation increase in IL leads to an increase of 1.14

[12] It is important to note that firm-specific weekly returns are not expected to be normally distributed. Prior literature (Kim, Li, and Zhang 2011b; Hutton, Marcus, and Tehranian 2009) uses such a cutoff point to obtain a reasonable and convenient benchmark for extreme events.



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in crash risk. Given the sample mean crash risk of 8.18, this effect is equivalent to an increase of crash risk by 14% relative to the mean, which is economically significant.

Overall, we find that companies with higher exposure to intangible liabilities have significantly higher future firm-specific crash risks. This result is robust across different measures of crash risk and holds across different specifications.

## 8. Analyzing different sections of the filings

The results presented so far are based on an IL measure constructed from the full text of annual filings. We previously argued that it is important to examine all sections together, as crucial information about intangible liabilities may be distributed across different sections. For example, in the Risk Factors section, a company might warn about potential harmful side effects of its products or about likely regulations that, if passed, would impose significant costs on the company. Additionally, in the Legal Proceedings and Commitments & Contingencies sections (when available), companies discuss past events (e.g., lawsuits) that are likely to generate future liabilities for the firm.

To assess whether the texts of both the Risk Factors section and the other sections possess predictive power for explaining stock returns, we construct two IL measures based on: (a) the Risk Factors section, and (b) the entire annual filing excluding the Risk Factors section. We then use these two measures in Fama-MacBeth regressions as signals for predicting stock returns.

Table 10 presents the results. As shown, both the Risk Factors section and the remaining sections of the filings contain relevant information related to intangible liabilities that can predict stock returns. However, the magnitude of the effect is smaller for the Risk Factors section compared to the rest of the text. Appendix Table 9 reports the time series abnormal returns of portfolios sorted based on the IL measure constructed from the Risk Factors section versus the one constructed from the other sections of the filings. Once again, both measures generate economically sizable and statistically significant abnormal returns across all six different factor models we study. In this time-series portfolio analysis, however, the excess returns are larger in magnitude for the Risk Factors section than for the rest of the filing. Therefore, while we do find significant effect of both



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IL measures on stock returns, our results on which measure has larger magnitude depends on particular empirical setup (Fama-MacBeth regressions versus time-series portfolio analysis).

Overall, our analysis demonstrates that the informational content of IL does not originate solely from the Risk Factors section, nor exclusively from the other sections of the filings. As a result, we conclude, as we initially argued, that it is essential to construct measures of IL based on all relevant information in the filings, regardless of the section in which it appears.

## 9. Risk versus mispricing

In the previous sections, we presented evidence indicating that companies with higher intangible liabilities (IL) tend to have higher abnormal returns. As discussed in the introduction, this result is expected theoretically, as investors would demand a higher return to compensate for the perceived risk of investing in such companies. An alternative hypothesis is that these companies are mispriced. However, our factor regression results, which control for mispricing factors (columns 5 and 6 in Panel B of Table 7), do not support this hypothesis. Specifically, the mispricing factors proposed by Stambaugh and Yuan (2016) and the long- and short-horizon behavioral mispricing factors of Daniel, Hirshleifer, and Sun (2019) do not explain the abnormal performance of companies with high IL. In fact, we find that the abnormal returns of the IL portfolio are even larger after controlling for these mispricing factors.

In this section, we further test the mispricing hypothesis in the context of earnings announcements. Specifically, it is possible that market participants overestimate the negative impact of intangible liabilities and, as a result, underestimate the future earnings of companies with high IL. If this were the case, these companies would exhibit higher earnings surprises, and the corresponding positive market reaction following the earnings announcements could explain the abnormal returns we have documented. To test this hypothesis, we examine whether companies with high IL experience larger positive earnings surprises and greater (abnormal) returns around their earnings announcement dates.

To test this mispricing hypothesis, we run regressions where the dependent variables are either measures of earnings surprises or cumulative abnormal returns around the announcement dates.



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We use two measures of earnings surprises: quarterly standardized unexpected earnings using a seasonal random walk model (SUE) and quarterly standardized unexpected earnings using median analyst forecasts (SUEAF). Abnormal returns are measured using the cumulative 3-day abnormal return around the earnings announcement (CAR). In our full models, we control for size, book-to-market ratio, and include both time and industry fixed effects.

The results, reported in Table 11, show no evidence that companies with high IL have higher earnings surprises, as indicated in columns 1-6. Additionally, the results in column 9 suggest that companies with high IL do not exhibit a positive market reaction around the earnings announcement dates. If anything, the abnormal returns around the announcement dates tend to be negative (although statistically insignificant).

Overall, while it is extremely challenging (if not impossible) to fully disentangle risk from mispricing in contexts similar to ours, our findings do not support the idea that mispricing plays a significant role in explaining the IL premium.

## 10. Identifying the Most Predictive Words

Our IL dictionary consists of more than 500 words, which has the advantage of capturing different languages and word usage by companies. As several hundreds of word frequencies contribute to the overall IL measure, one useful task is determining which specific words are most important in driving the observed relationship between IL and future returns in our sample. Traditional ordinary least squares (OLS) regression struggles with multicollinearity and overfitting when the number of predictors is large. To address this issue, we employ the Least Absolute Shrinkage and Selection Operator (LASSO) regression, which performs variable selection and regularization simultaneously. LASSO imposes an L1 penalty (or LASSO penalty) that forces some coefficients to zero, thereby identifying the most relevant predictors while mitigating overfitting. This approach is particularly useful in our setup, as it allows us to determine the subset of words that contribute most significantly to the IL and stock return predictability.

To implement LASSO regression, we first standardize the frequency of each litigious word to ensure comparability across predictors. Next, we divide the sample into a training set (pre-2015) and a test set (post-2015) to assess predictive performance. Our dependent variable is the one month



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ahead stock return and the independent variables are standardized frequencies of individual words in addition to BETA, SIZE, BM, and MOM. The LASSO model is then estimated using multiple penalty values, allowing the regularization strength to vary and enabling optimal variable selection. We consider lambda values of 0.1, 0.05, 0.02, and 0.01, and select the model where lambda minimizes cross-validation error. The optimal lambda identified through this process is 0.02. Finally, we examine the coefficients to identify the most significant predictors based on their standardized values and evaluate the predictive power of the selected model on the test dataset to ensure robustness and out-of-sample validity.

The LASSO regression identifies 149 most important words, indicating their significance in predicting stock returns. The sorted standardized coefficients of individual words reveal that the top most important predictors are related to litigation and legal disputes (*infringement*, *fraud*, *mediated*, *prosecution*) as well as words related to contracts, business regulations and legal frameworks (*nullifies*, *product liability*, *promulgated*, *rescission*, *jurisdictions*, *enforceability*). Examples of other commonly used words that show up important are: *litigation*, *settlement*, *appeals*, *restitution*, *contingency*, *indemnifications*, and *forfeitable*, which are also related to legal proceedings, resolving disputes and seeking remedies in court. We list the full set of 149 words in Appendix Table 9.

While the Lasso regression identifies 149 words with non-zero coefficients as the most predictive features of stock returns, it is important to emphasize that this does not imply the remaining words are irrelevant or lack predictive power. Lasso is designed to perform both coefficient shrinkage and variable selection by imposing a penalty on the absolute size of the coefficients. In doing so, it applies a relatively harsh regularization constraint that forces many coefficients exactly to zero — even when some excluded variables may still carry informative signals.

The fact that a coefficient is zero, therefore, does not imply that the corresponding word has no association with stock returns. Rather, it reflects the tradeoff Lasso makes between model complexity and predictive performance. Many of the excluded words may still be informative, particularly if they are highly correlated with selected words or if their marginal contribution is modest but meaningful. In such cases, Lasso may assign a zero coefficient not because the word is unrelated to the outcome, but because its predictive signal is redundant or overshadowed by



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stronger, correlated predictors. Moreover, the set of selected words depends on the level of regularization imposed, and different penalty choices may yield different subsets. Thus, while the 149 selected words represent the most prominent predictors under the Lasso framework, the remaining words should not be interpreted as entirely uninformative or irrelevant.

## 11. Conclusion

A growing literature studies the characteristics and valuation of companies with a high level of intangible assets. While difficult to precisely define, the common feature of intangible assets is that they are economically important but are not reported on the balance sheet. Following this concept, in this paper we make a first attempt at measuring intangible liabilities. These are liabilities that, while economically important, do not show up on the balance sheet. We propose a text-based measure of intangible liabilities (IL) and show that it varies in a meaningful way across industries and firms. Our measure is informative about future class action lawsuits against firms and predicts future deterioration in firms' reputation. We also find that firms with higher exposure to IL are traded at lower valuation ratios, and investors appear to demand higher expected returns to invest in these companies. Consistently, we show that higher exposure to IL predicts future firm-specific crash risk.

Like all proposed measures of intangible *assets*, our measure of intangible *liabilities* is not without its limitations. One key area for improvement is the incorporation of additional sources, such as information from earnings calls, social media, and press coverage, to develop a more comprehensive measure of intangible liabilities. Another avenue worth exploring is whether more advanced natural language processing techniques could generate more informative measures. We view our paper as a first step toward emphasizing and drawing attention to the significance of intangible liabilities. Given the uncertain and contingent nature of these liabilities, they are likely to impact critical firm-level outcomes, such as financial policies, hiring and firing practices, target selection and premiums in acquisitions, and labor/CEO compensation contracts, among others. We believe these topics present valuable opportunities for future research.



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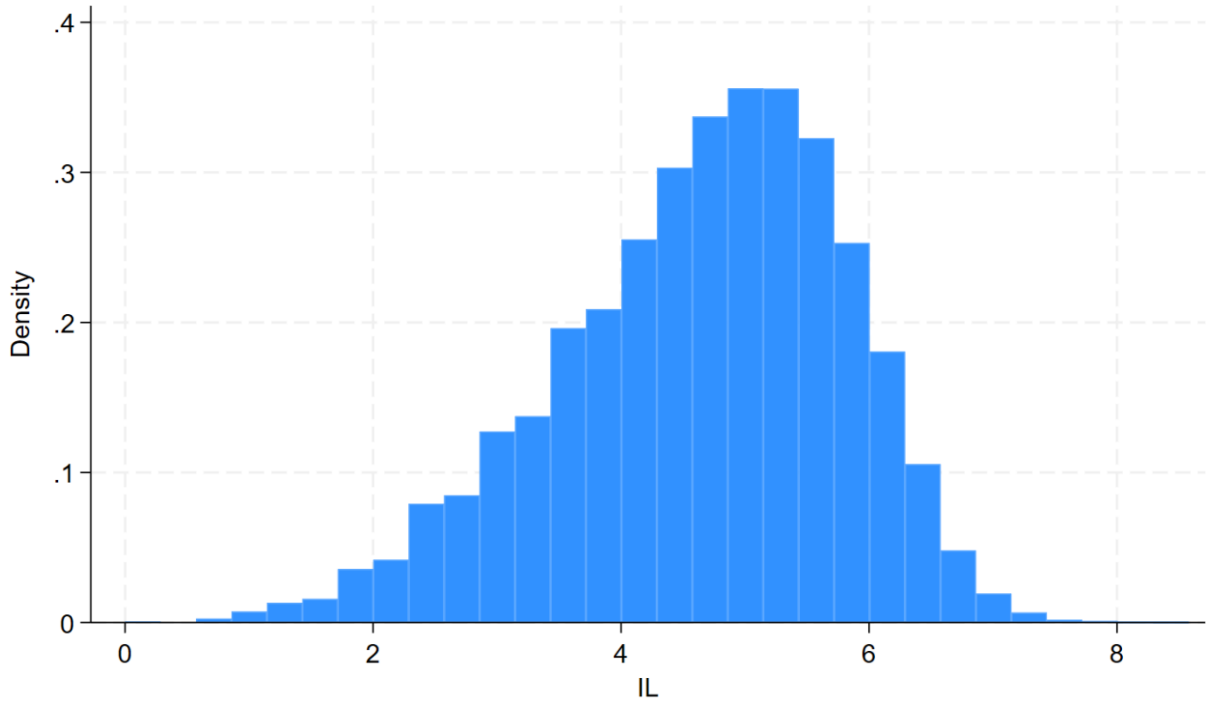
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Figure 1. Histogram of the text-based intangible liability measure

This figure displays the histogram of the logarithm of word frequency from the intangible liability dictionary, *IL*, in the annual filings of companies. The sample is from fiscal years 1996 to 2020. The list of words in the dictionary is provided in Appendix Table 1.

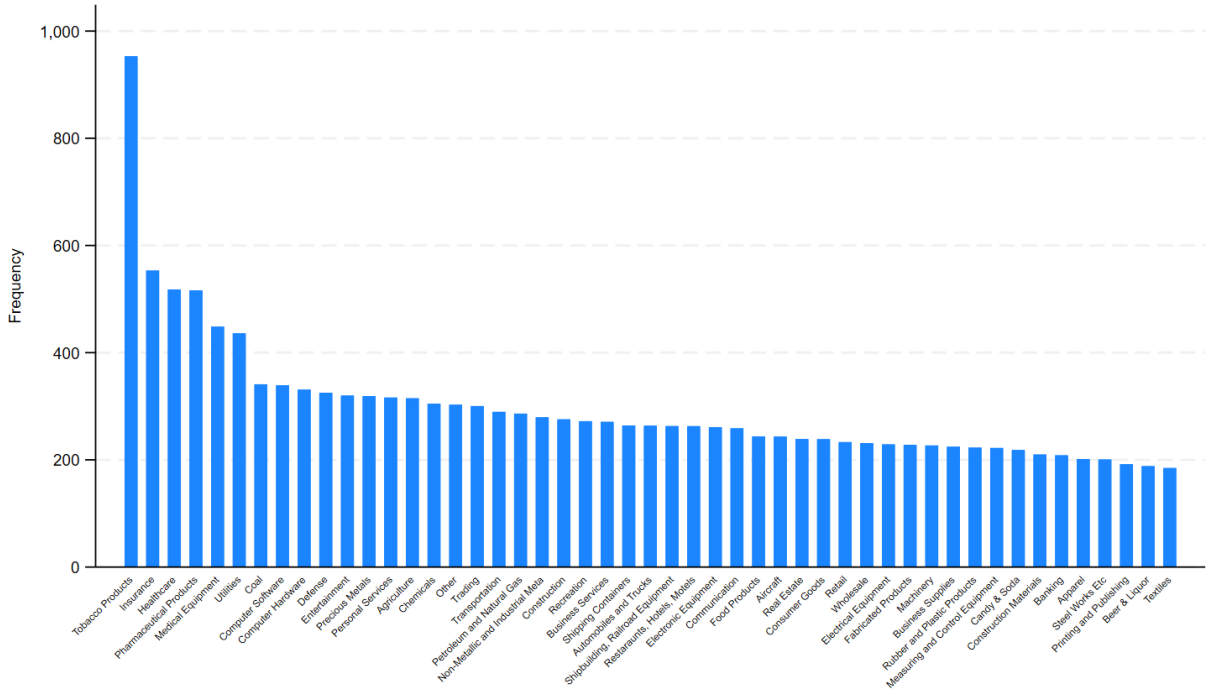





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Figure 2. Intangible liabilities, by industry

The figure plots the average word frequency from intangible liabilities dictionary in the annual filings of companies, grouped by Fama-French 49 industry classification.

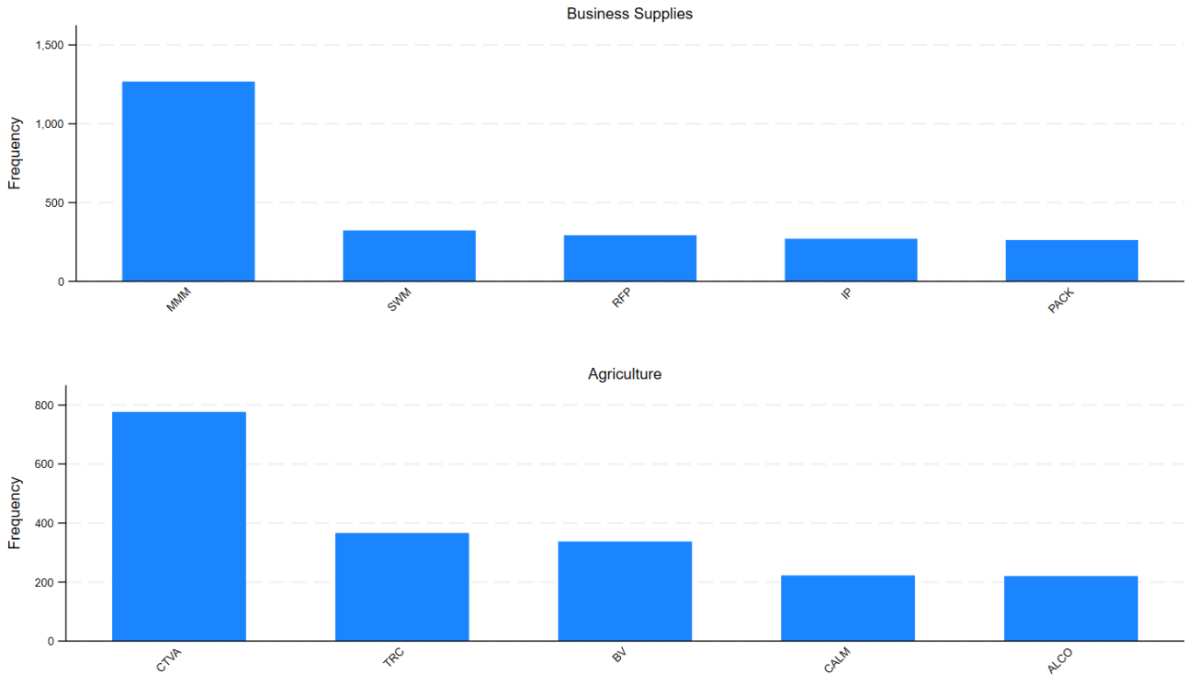





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Figure 3. Top 5 companies with highest intangible liabilities in two selected industries

This figure plots the frequency of intangible liabilities words in fiscal year 2020 for the top 5 companies in two industries: Business Supplies and Agriculture.

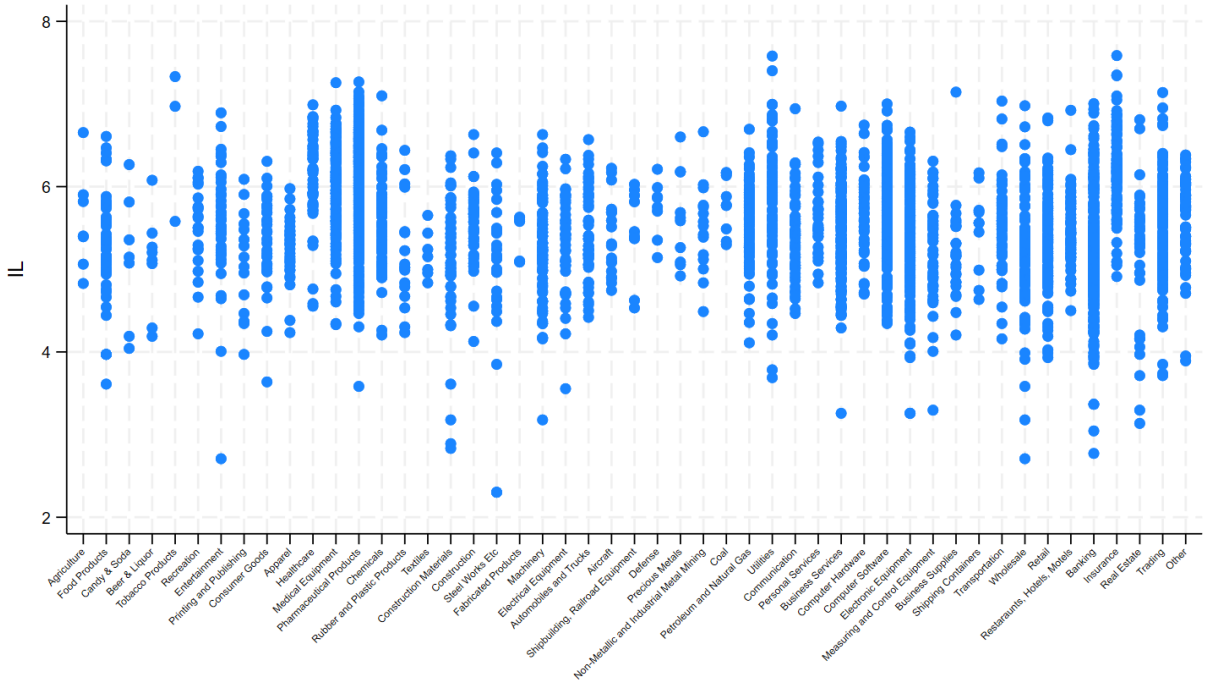





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Figure 4. Within industry distribution of intangible liabilities

This figure shows the distribution of intangible liabilities measure, *IL*, within each of Fama-French 49 industry classification in fiscal year 2020. Each dot represents one company. *IL* is the logarithm of word frequency from the intangible liability dictionary in the annual filings of companies.

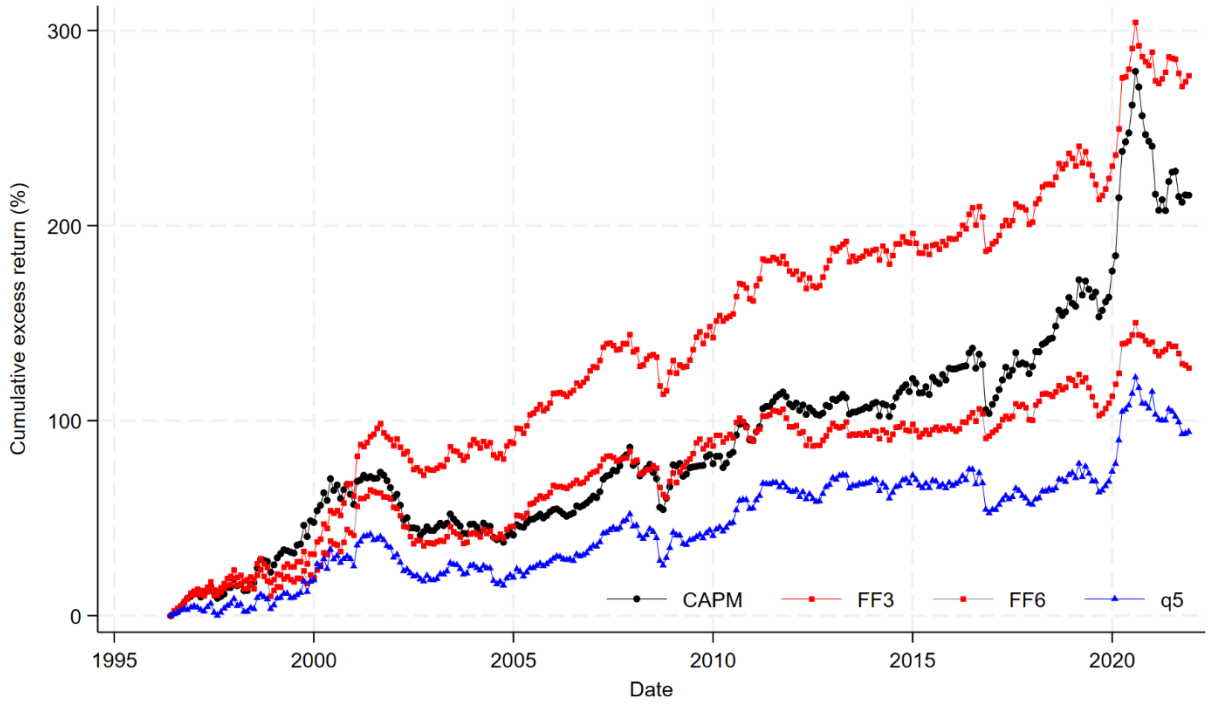





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Figure 5. Cumulative excess return of IL long-short portfolio

This figure plots the cumulative excess return (alpha plus residuals) of a portfolio that is long (short) the top (bottom) quintile of companies sorted based on intangible liabilities, IL. We show the excess returns using four factor models: CAPM, Fama-French 3 factor (FF3), Fama-French 5-factor plus momentum (FF6) and the q5 model.






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Table 1. Descriptive statistics

This table presents descriptive statistics on the variables used in our main tests. This sample consists of all firm-year observations for which we could download the 10-K annual reports and find a match in Compustat and CRSP. Sample is from 1996 to 2020. Variable definitions are reported in Appendix Table 2.

| | N | Mean | SD | Min | p50 | Max |
|---|---|---|---|---|---|---|
| IL | 112,457 | 4.64 | 1.15 | 0.00 | 4.79 | 8.58 |
| UNIQUE | 112,457 | 39.30 | 23.16 | 1.00 | 36.00 | 149.00 |
| PE | 73,360 | 33.65 | 63.04 | 2.21 | 17.82 | 497.85 |
| MB | 106,916 | 3.45 | 4.81 | 0.26 | 1.98 | 34.10 |
| SIZE | 112,442 | 12.65 | 2.20 | 4.39 | 12.57 | 21.79 |
| ROA | 111,037 | -0.02 | 0.27 | -1.50 | 0.05 | 0.31 |
| LEVERAGE | 112,079 | 0.17 | 0.20 | 0.00 | 0.10 | 0.90 |
| TANGIBILITY | 97,661 | 0.46 | 0.40 | 0.00 | 0.34 | 1.83 |
| TOTAL WORDS | 112,457 | 10.00 | 0.74 | 7.60 | 10.16 | 13.59 |
| ABNORMAL RET | 112,457 | 0.03 | 0.62 | -1.00 | -0.05 | 3.01 |
| VOLATILITY | 111,404 | 0.15 | 0.10 | 0.03 | 0.12 | 0.58 |
| TURNOVER | 102,457 | 0.20 | 0.24 | 0.01 | 0.13 | 1.51 |
| TOBIN Q | 110,602 | 1.81 | 2.16 | 0.05 | 1.12 | 13.29 |
| LAWSUIT | 105,856 | 0.02 | 0.15 | 0.00 | 0.00 | 1.00 |
| RRI | 13,042 | 10.42 | 12.32 | 0.00 | 5.00 | 70.00 |



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Table 2. Descriptive statistics, grouped by IL quintiles.

This table provides the average value of each variable within each quintile of intangible liabilities, IL. Panel A shows these values for our main sample (1996-2020), whereas Panel B reports similar statistics for the restricted Reprisk sample. IL is measured by the log frequency of words in the IL dictionary. All variables are defined in Appendix Table 2.

Panel A: Core sample

| | IL Quintile | | | | |
|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 |
| IL | 3.38 | 4.19 | 4.67 | 5.14 | 5.88 |
| UNIQUE | 18.45 | 28.82 | 37.17 | 47.21 | 65.38 |
| PE | 28.21 | 31.70 | 37.12 | 38.21 | 35.25 |
| MB | 2.63 | 3.17 | 3.74 | 3.94 | 3.82 |
| TOBIN Q | 1.39 | 1.68 | 2.01 | 2.11 | 1.88 |
| SIZE | 11.92 | 12.37 | 12.66 | 12.90 | 13.42 |
| ROA (%) | 2.78 | 0.00 | -3.26 | -5.87 | -5.69 |
| LEVERAGE | 0.13 | 0.16 | 0.18 | 0.18 | 0.20 |
| TANGIBILITY | 0.52 | 0.49 | 0.46 | 0.42 | 0.41 |
| TOTAL WORDS | 9.47 | 9.89 | 10.04 | 10.18 | 10.46 |
| ABNORMAL RET | 0.03 | 0.04 | 0.03 | 0.03 | 0.01 |
| VOLATILITY | 0.12 | 0.14 | 0.15 | 0.16 | 0.16 |
| TURNOVER | 0.12 | 0.17 | 0.22 | 0.24 | 0.26 |
| LAWSUIT (%) | 0.32 | 0.71 | 1.53 | 3.50 | 5.66 |

Panel B: RepRisk sample

| | IL Quintile | | | | |
|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 |
| IL | 4.54 | 5.12 | 5.48 | 5.84 | 6.41 |
| UNIQUE | 33.55 | 45.39 | 54.95 | 66.24 | 83.73 |
| TOBIN Q | 1.78 | 1.73 | 1.83 | 1.92 | 1.84 |
| SIZE | 14.22 | 14.53 | 14.69 | 15.02 | 15.35 |
| ROA | 7.09 | 6.31 | 5.14 | 3.54 | 0.21 |
| LEVERAGE | 0.22 | 0.25 | 0.26 | 0.27 | 0.26 |
| TANGIBILITY | 0.60 | 0.55 | 0.52 | 0.48 | 0.39 |
| TOTAL WORDS | 10.10 | 10.44 | 10.58 | 10.72 | 10.95 |
| ABNORMAL RET | 0.02 | 0.02 | 0.03 | 0.04 | 0.03 |
| VOLATILITY | 0.11 | 0.11 | 0.11 | 0.12 | 0.11 |
| TURNOVER | 0.21 | 0.24 | 0.27 | 0.29 | 0.28 |
| RRI (%) | 8.45 | 7.90 | 9.09 | 11.25 | 14.34 |



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Table 3. Intangible liabilities and other firm characteristics

This table shows the result of regressions of intangible liabilities, IL, on firm characteristics. IL is measured by the log frequency of words in the IL dictionary that show up in each firm's annual filing. Columns 1-3 include industry fixed effects whereas column 4 controls for firm fixed effects. All specifications include year fixed effects. Standard errors, clustered at the industry level in columns 1-3 and at the firm level in column 4, are shown in parentheses. All variables are defined in Appendix Table 2. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Dep. Var.: | | | IL | |
| SIZE | 0.10*** | 0.04*** | 0.05*** | -0.00 |
| | (0.01) | (0.00) | (0.00) | (0.00) |
| ROA | | -0.10** | -0.11*** | -0.04*** |
| | | (0.04) | (0.03) | (0.01) |
| LEVERAGE | | -0.10** | -0.11*** | -0.08*** |
| | | (0.04) | (0.04) | (0.02) |
| TANGIBILITY | | 0.00 | -0.01 | 0.04** |
| | | (0.02) | (0.02) | (0.02) |
| TOTAL WORDS | | 1.23*** | 1.20*** | 1.12*** |
| | | (0.02) | (0.02) | (0.01) |
| ABNORMAL RET | | | -0.03*** | -0.00 |
| | | | (0.01) | (0.00) |
| VOLATILITY | | | 0.30*** | 0.08*** |
| | | | (0.07) | (0.03) |
| TURNOVER | | | 0.06*** | 0.02* |
| | | | (0.02) | (0.01) |
| TOBIN Q | | | -0.02*** | -0.02*** |
| | | | (0.00) | (0.00) |
| Observations | 112,442 | 96,999 | 87,161 | 87,161 |
| Adjusted $R^2$ | 0.51 | 0.75 | 0.74 | 0.70 |
| Industry FE | Yes | Yes | Yes | Implied |
| Year FE | Yes | Yes | Yes | Yes |
| Firm FE | No | No | No | Yes |



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Table 4. Intangible liabilities and valuation ratios

This table shows results of OLS regressions of valuation ratios on intangible liabilities, *IL*, and firm level control variables. The dependent variable in the first two columns is the natural logarithm of price-to-earnings ratio (PE), whereas it is the natural logarithm of market-to-book (MB) ratio in columns 3-4. IL is the logarithm of frequency of words in the IL dictionary that shows up in each firm's annual filing. Columns 1 and 3 include industry effects, whereas columns 2 and 4 include firm fixed effects. All specifications include year fixed effects. Standard errors, clustered at the industry level in columns 1 and 3 and at the firm level in columns 2 and 4, are shown in parentheses. All variables are defined in Appendix Table 2. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Dep. Var.: | LOG PE | | LOG MB | |
| IL | -0.04** | -0.03*** | -0.05*** | -0.05*** |
| | (0.02) | (0.01) | (0.02) | (0.01) |
| SIZE | 0.11*** | 0.41*** | 0.23*** | 0.48*** |
| | (0.01) | (0.01) | (0.01) | (0.01) |
| LEVERAGE | -0.65*** | -0.33*** | 0.29*** | 0.73*** |
| | (0.07) | (0.05) | (0.09) | (0.04) |
| ROA | -4.91*** | -8.79*** | -0.58*** | -0.44*** |
| | (0.39) | (0.33) | (0.10) | (0.02) |
| VOLATILITY | 1.34*** | 0.88*** | 1.42*** | 0.80*** |
| | (0.13) | (0.08) | (0.11) | (0.04) |
| TANGIBILITY | 0.23*** | 0.91*** | -0.05 | 0.60*** |
| | (0.04) | (0.05) | (0.04) | (0.03) |
| TOTAL WORDS | 0.01 | 0.06*** | -0.13*** | -0.04*** |
| | (0.03) | (0.02) | (0.02) | (0.01) |
| Observations | 59,431 | 59,431 | 90,798 | 90,798 |
| R-squared | 0.22 | 0.34 | 0.34 | 0.39 |
| Industry FE | Yes | Implied | Yes | Implied |
| Year FE | Yes | Yes | Yes | Yes |
| Firm FE | No | Yes | No | Yes |



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Table 5. Intangible liabilities and future lawsuits

This table shows the relationship between intangible liabilities, IL, and future lawsuits. The dependent variable is an indicator equal to 1 if the firm has experienced a lawsuit in year t+1 and 0 otherwise. Panel A contains results from logit specifications, whereas Panel B shows results from linear probability models. Each Panel A column/specification has the same control variables as the corresponding one in Panel B, but the estimated coefficients are dropped in Panel A for brevity. IL is measured by natural logarithm of frequency of words in the IL dictionary that shows up in each firm's annual filing. Standard errors are shown in parentheses. Columns 1-4 include industry effects, whereas columns 5-8 have firm fixed effects. All specifications include year effects. All variables are defined in Appendix Table 2. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Dep variable: | Lawsuit (t+1) | | | | | | | |
| | Panel A. Logit Regressions | | | | | | | |
| IL | 1.14*** | 1.08*** | 0.86*** | 1.03*** | 1.23*** | 1.22*** | 1.13*** | 1.37*** |
| | (0.05) | (0.05) | (0.05) | (0.06) | (0.05) | (0.05) | (0.06) | (0.08) |
| | Panel B. Linear Probability Model | | | | | | | |
| IL | 2.10*** | 1.97*** | 1.47*** | 1.99*** | 2.39*** | 2.39*** | 2.04*** | 2.57*** |
| | (0.19) | (0.18) | (0.15) | (0.23) | (0.11) | (0.11) | (0.11) | (0.15) |
| SIZE | | 0.23*** | 0.56*** | 0.58*** | | -0.33*** | 0.55*** | 0.58*** |
| | | (0.05) | (0.08) | (0.09) | | (0.07) | (0.09) | (0.09) |
| ABNORMAL RET | | | -2.83*** | -2.85*** | | | -2.77*** | -2.79*** |
| | | | (0.20) | (0.20) | | | (0.12) | (0.12) |
| VOLATILITY | | | 15.61*** | 16.04*** | | | 16.93*** | 17.18*** |
| | | | (1.34) | (1.35) | | | (1.10) | (1.11) |
| TURNOVER | | | 6.12*** | 6.21*** | | | 6.96*** | 7.00*** |
| | | | (0.51) | (0.51) | | | (0.52) | (0.52) |
| LEVERAGE | | | -0.58 | -0.24 | | | -0.57 | -0.39 |
| | | | (0.44) | (0.46) | | | (0.50) | (0.50) |
| ROA | | | 0.11 | -0.00 | | | -2.64*** | -2.75*** |
| | | | (0.63) | (0.62) | | | (0.47) | (0.47) |
| TANGIBILITY | | | -0.49* | -0.61** | | | -1.48*** | -1.61*** |
| | | | (0.27) | (0.27) | | | (0.34) | (0.34) |
| TOTAL WORDS | | | | -1.35*** | | | | -1.47*** |
| | | | | (0.29) | | | | (0.24) |
| Observations | 105,856 | 105,843 | 85,758 | 85,758 | 105,856 | 105,843 | 85,758 | 85,758 |
| R-squared | 0.02 | 0.02 | 0.05 | 0.05 | 0.01 | 0.01 | 0.04 | 0.04 |
| Industry FE | Yes | Yes | Yes | Yes | Implied | Implied | Implied | Implied |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Firm FE | No | No | No | No | Yes | Yes | Yes | Yes |



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Table 6. Intangible liabilities and future reputation risk

This table shows the relationship between intangible liabilities, IL, and future reputation risk, RRI, provided by Reprisk. Reputation risk, the dependent variable, is a variable ranging between 0 and 100% with higher values indicating higher reputational risk as measured by negative media attention in the next year. IL is measured by natural logarithm of frequency of words in the IL dictionary that shows up in each firm's annual filing. Columns 1-4 include industry effects, while columns 5-8 include firm fixed effects. All specifications include year effects. Standard errors, clustered at the industry level in columns 1-4 and at the firm level in columns 5-8, are shown in parentheses. All variables are defined in Appendix Table 2. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Dep variable: | RRI (t+1) | | | | | | | |
| IL | 4.69*** | 2.25*** | 1.88*** | 2.66*** | 1.29*** | 1.29*** | 1.07*** | 0.99** |
| | (0.41) | (0.31) | (0.35) | (0.55) | (0.37) | (0.37) | (0.39) | (0.51) |
| SIZE | | 3.36*** | 3.92*** | 3.95*** | | 0.90*** | 1.35*** | 1.35*** |
| | | (0.20) | (0.22) | (0.21) | | (0.17) | (0.22) | (0.22) |
| ABNORMAL RET | | | -2.31*** | -2.33*** | | | -0.71*** | -0.71*** |
| | | | (0.19) | (0.19) | | | (0.20) | (0.20) |
| VOLATILITY | | | 14.27*** | 14.53*** | | | -1.15 | -1.16 |
| | | | (2.93) | (2.87) | | | (1.88) | (1.88) |
| TURNOVER | | | 0.04 | 0.15 | | | 1.43** | 1.43** |
| | | | (0.88) | (0.91) | | | (0.61) | (0.61) |
| LEVERAGE | | | -1.09 | -0.60 | | | 0.91 | 0.88 |
| | | | (0.84) | (0.78) | | | (0.94) | (0.95) |
| ROA | | | -6.39** | -6.67*** | | | -1.97** | -1.95** |
| | | | (2.43) | (2.41) | | | (0.82) | (0.83) |
| TANGIBILITY | | | 2.73*** | 2.66*** | | | 0.95 | 0.98 |
| | | | (0.65) | (0.64) | | | (0.91) | (0.91) |
| TOTAL WORDS | | | | -1.85 | | | | 0.22 |
| | | | | (1.22) | | | | (0.99) |
| Observations | 12,598 | 12,598 | 11,256 | 11,256 | 12,598 | 12,598 | 11,256 | 11,256 |
| R-squared | 0.11 | 0.36 | 0.39 | 0.39 | 0.01 | 0.01 | 0.02 | 0.02 |
| Industry FE | Yes | Yes | Yes | Yes | Implied | Implied | Implied | Implied |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Firm FE | No | No | No | No | Yes | Yes | Yes | Yes |



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Table 7. Intangible liabilities and stock returns: Time-series portfolio analysis

This table shows results of time-series portfolios (monthly frequency) constructed based on the intangible liabilities, IL. Panel A shows the annualized average returns and industry-adjusted returns of five portfolios sorted based on IL. Panel B reports the annualized excess returns or alpha of the long-short IL portfolio (highest quintile minus lowest quintile) using different factor models: CAPM, Fama-French 3-factor (FF3), Fama-French 5-factor plus momentum [FF6], Hou et al. (2020) [q5], Daniel, Hirshleifer, and Sun (2019) [DHS], and Stambaugh and Yuan (2016) [SY]. The *t*-statistic computed using Newey-West standard errors are reported in parentheses. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Panel A. Portfolio returns | | | | | | |
| IL: | Low | 2 | 3 | 4 | High | High - Low |
| Average raw return | 5.68* | 8.13** | 6.89* | 7.79** | 10.03*** | 4.35** |
| | (1.67) | (2.40) | (1.92) | (2.02) | (3.06) | (2.49) |
| Ind-adj return | 7.15** | 7.50** | 7.26** | 7.26** | 10.17*** | 3.02** |
| | (2.34) | (2.19) | (2.10) | (2.02) | (2.98) | (2.19) |
| Panel B. Portfolio abnormal return, alphas (high minus low IL) | | | | | | |
| | CAPM | FF3 | FF6 | q5 | DHS | SY |
| α (%) | 4.89*** | 5.47*** | 3.45*** | 2.91** | 5.77*** | 6.02*** |
| | (2.57) | (3.24) | (2.72) | (2.07) | (3.96) | (2.70) |
| Ind-adj α (%) | 3.09** | 3.39** | 2.99** | 2.11* | 3.48*** | 3.16** |
| | (2.24) | (2.36) | (2.43) | (1.71) | (2.64) | (2.04) |



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Table 8. Intangible liabilities and stock returns: Fama-MacBeth regressions

This table reports the results of predictive month by month cross-sectional regressions using Fama-MacBeth methodology. The coefficients reported are the average time-series of cross-sectional estimates. All variables are standardized to have a mean of 0 and standard deviation of 1. The *t*-statistic computed using Newey-West standard errors are reported in parentheses. Variable definitions are reported in Appendix Table 2. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dep var: | | | Returns (t+1) | | | |
| IL | 0.17** | 0.18*** | 0.17*** | 0.16*** | 0.13*** | 0.16*** |
| | (2.37) | (2.61) | (2.83) | (3.12) | (2.78) | (4.17) |
| BETA | 0.05 | 0.04 | 0.10 | 0.10 | 0.05 | 0.02 |
| | (0.31) | (0.27) | (0.81) | (1.18) | (0.61) | (0.25) |
| SIZE | -0.10 | -0.11 | -0.17 | -0.32* | -0.21 | -0.19 |
| | (-0.64) | (-0.75) | (-1.44) | (-1.93) | (-1.38) | (-1.27) |
| BM | 0.18 | 0.29** | 0.21** | 0.12 | 0.10 | 0.09 |
| | (1.35) | (2.27) | (2.00) | (1.19) | (1.05) | (0.81) |
| MOM | | 0.29*** | 0.24*** | 0.25*** | 0.30*** | 0.26*** |
| | | (3.22) | (2.94) | (3.07) | (3.70) | (3.22) |
| PROFIT | | | 0.17* | 0.14* | 0.26*** | 0.18** |
| | | | (1.82) | (1.82) | (3.85) | (2.50) |
| INV | | | -0.29*** | -0.23*** | -0.17*** | -0.10* |
| | | | (-5.14) | (-3.98) | (-3.31) | (-1.78) |
| VOL | | | | 0.33 | 0.21 | 0.21 |
| | | | | (1.52) | (1.02) | (0.97) |
| IDIOVOL | | | | -0.52*** | -0.55*** | -0.48*** |
| | | | | (-5.30) | (-5.72) | (-4.76) |
| SKEW | | | | -0.06* | -0.06 | -0.04 |
| | | | | (-1.69) | (-1.57) | (-0.96) |
| ILLIQ | | | | 0.24*** | 0.27*** | 0.31*** |
| | | | | (4.99) | (5.35) | (6.42) |
| PRICE | | | | -0.47*** | -0.35** | -0.35* |
| | | | | (-2.66) | (-2.15) | (-1.96) |
| YIELD | | | | 0.13*** | 0.18*** | 0.17** |
| | | | | (3.04) | (3.87) | (2.33) |
| AdEXP | | | | | 0.02 | 0.03 |
| | | | | | (0.42) | (0.60) |
| R&D | | | | | 0.60*** | 0.57*** |
| | | | | | (5.92) | (5.94) |
| LEVERAGE | | | | | | -0.02 |
| | | | | | | (-0.25) |
| ACCRUALS | | | | | | -0.06 |
| | | | | | | (-1.59) |
| REPURCHASE | | | | | | 0.02 |
| | | | | | | (0.59) |
| NETEQUITY | | | | | | -0.04 |
| | | | | | | (-0.79) |
| TOTAL WORDS | -0.18*** | -0.18*** | -0.14*** | -0.14*** | -0.12*** | -0.17*** |
| | (-4.30) | (-4.39) | (-3.16) | (-3.15) | (-2.96) | (-3.83) |
| Observations | 1,096,754 | 1,095,767 | 1,065,287 | 1,064,513 | 1,064,513 | 762,477 |
| R-squared | 0.04 | 0.04 | 0.05 | 0.06 | 0.07 | 0.07 |



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Table 9. Intangible liabilities and future firm-specific crash risk

This table shows results on the relation between intangible liabilities, IL, and future firm-specific crash risk. We use three measures of crash risk as dependent variables. The difference between specifications within each measure of crash risk is the number of control variables used. All regressions include year and industry fixed effects as well as total file words. The *t*-statistic computed using clustered standard errors at the firm level are reported in parentheses. Variable definitions are reported in Appendix Table 2. ***, **, * denote significance at 1%, 5%, and 10%, respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| DEP VAR: | NCSKEW (t+1) | | | CRASH (t+1) | | | CRASH_EXTR (t+1) | | |
| IL | 3.16*** | 0.79** | 0.98** | 1.46*** | 0.97*** | 0.88*** | 3.48*** | 1.92*** | 1.99*** |
| | (6.20) | (2.04) | (2.19) | (6.75) | (4.12) | (3.21) | (7.98) | (4.08) | (3.63) |
| NCSKEW | | 0.02*** | 0.02*** | | | | | | |
| | | (5.04) | (4.06) | | | | | | |
| CRASH | | | | | 0.03*** | 0.03*** | | | |
| | | | | | (8.88) | (7.78) | | | |
| CRASH_EXTR | | | | | | | | 0.03*** | 0.03*** |
| | | | | | | | | (8.57) | (7.14) |
| DTURN | | 22.42*** | 23.39*** | | 7.39*** | 8.50*** | | 21.10*** | 23.22*** |
| | | (6.98) | (6.41) | | (5.54) | (5.63) | | (7.74) | (7.58) |
| RET | | 34.78*** | 32.48*** | | 6.54*** | 6.15*** | | 22.33*** | 20.79*** |
| | | (10.96) | (8.53) | | (5.13) | (4.00) | | (8.26) | (6.36) |
| SIGMA | | 243.65*** | 202.20*** | | 33.16** | 22.92 | | 133.14*** | 102.26*** |
| | | (7.36) | (5.00) | | (2.34) | (1.31) | | (4.59) | (2.87) |
| SIZE | | 4.95*** | 5.08*** | | 0.42*** | 0.52*** | | 1.37*** | 1.70*** |
| | | (21.75) | (17.93) | | (3.90) | (3.88) | | (6.62) | (6.57) |
| MB | | 11.49*** | 11.04*** | | 2.44*** | 2.33*** | | 6.61*** | 6.39*** |
| | | (28.03) | (23.63) | | (13.62) | (11.31) | | (18.24) | (15.37) |
| LEV | | -15.28*** | -17.97*** | | -2.84*** | -4.12*** | | -5.77*** | -9.74*** |
| | | (-7.14) | (-7.13) | | (-2.96) | (-3.68) | | (-2.97) | (-4.27) |
| ROA | | 13.37*** | 12.19*** | | 2.88*** | 2.84*** | | 8.24*** | 8.07*** |
| | | (6.80) | (5.57) | | (3.48) | (3.00) | | (4.68) | (4.07) |
| ACCM | | | 2.28 | | | 2.08** | | | 5.48*** |
| | | | (0.99) | | | (2.08) | | | (2.74) |
| Observations | 99,624 | 83,032 | 60,357 | 99,624 | 83,032 | 60,357 | 99,624 | 83,032 | 60,357 |
| R-squared | 0.02 | 0.05 | 0.05 | 0.02 | 0.03 | 0.03 | 0.03 | 0.04 | 0.04 |



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Table 10. Intangible liabilities and stock returns: Analyzing different sections of 10Ks

This table reports the results of predictive month by month cross-sectional regressions using Fama-MacBeth methodology similar as Table 8 with the difference that here we have two measures of intangible liabilities, IL, constructed from either the Risk Factor section or the rest of the filing. The coefficients reported are the average time-series of cross-sectional estimates. All variables are standardized to have a mean of 0 and standard deviation of 1. The *t*-statistic computed using Newey-West standard errors are reported in parentheses. Variable definitions are reported in Appendix Table 2. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dep var: | | | | Returns (t+1) | | |
| IL (Risk Factors) | 0.07* | 0.08** | 0.09** | 0.09*** | 0.06** | 0.08*** |
| | (1.94) | (2.18) | (2.50) | (3.30) | (2.49) | (2.67) |
| IL (Exc Risk Factors) | 0.15*** | 0.16*** | 0.14*** | 0.12*** | 0.11** | 0.14*** |
| | (2.95) | (3.18) | (2.90) | (2.61) | (2.54) | (3.66) |
| BETA | 0.06 | 0.04 | 0.11 | 0.10 | 0.05 | 0.01 |
| | (0.37) | (0.30) | (0.83) | (1.15) | (0.58) | (0.18) |
| SIZE | -0.12 | -0.13 | -0.18 | -0.32* | -0.21 | -0.19 |
| | (-0.79) | (-0.90) | (-1.57) | (-1.90) | (-1.35) | (-1.21) |
| BM | 0.18 | 0.27** | 0.20* | 0.11 | 0.10 | 0.08 |
| | (1.27) | (2.13) | (1.92) | (1.11) | (0.97) | (0.74) |
| MOM | | 0.28*** | 0.23*** | 0.24*** | 0.30*** | 0.26*** |
| | | (3.05) | (2.78) | (2.96) | (3.59) | (3.08) |
| PROFIT | | | 0.16* | 0.13* | 0.26*** | 0.18** |
| | | | (1.66) | (1.72) | (3.74) | (2.40) |
| INV | | | -0.29*** | -0.24*** | -0.17*** | -0.10* |
| | | | (-5.11) | (-4.01) | (-3.32) | (-1.80) |
| VOL | | | | 0.34 | 0.21 | 0.21 |
| | | | | (1.56) | (1.05) | (0.97) |
| IDIOVOL | | | | -0.51*** | -0.54*** | -0.47*** |
| | | | | (-5.15) | (-5.57) | (-4.61) |
| SKEW | | | | -0.06* | -0.06 | -0.04 |
| | | | | (-1.66) | (-1.54) | (-0.94) |
| ILLIQ | | | | 0.24*** | 0.27*** | 0.31*** |
| | | | | (4.94) | (5.27) | (6.37) |
| PRICE | | | | -0.50*** | -0.38** | -0.37** |
| | | | | (-2.81) | (-2.29) | (-2.08) |
| YIELD | | | | 0.12*** | 0.17*** | 0.16** |
| | | | | (2.92) | (3.74) | (2.21) |
| AdEXP | | | | | 0.03 | 0.03 |
| | | | | | (0.48) | (0.64) |
| R&D | | | | | 0.61*** | 0.58*** |
| | | | | | (5.95) | (5.96) |
| LEVERAGE | | | | | | -0.03 |
| | | | | | | (-0.30) |
| ACCRUALS | | | | | | -0.06 |
| | | | | | | (-1.64) |
| REPURCHASE | | | | | | 0.02 |
| | | | | | | (0.62) |
| NETEQUITY | | | | | | -0.04 |
| | | | | | | (-0.70) |
| TOTAL WORDS | -0.18*** | -0.18*** | -0.13*** | -0.13*** | -0.12*** | -0.16*** |
| | (-4.50) | (-4.56) | (-3.31) | (-3.17) | (-3.06) | (-3.73) |
| Observations | 1,085,055 | 1,084,071 | 1,053,693 | 1,052,968 | 1,052,968 | 753,657 |
| R-squared | 0.04 | 0.04 | 0.05 | 0.06 | 0.07 | 0.07 |



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Table 11. Earnings surprises and announcement returns

The table studies whether companies with high intangible liabilities, IL, have significantly higher earnings surprises and earnings announcement returns. Dependent variables are quarterly standardized unexpected earnings using a seasonal random walk model (SUE) in columns 1-3, quarterly standardized unexpected earnings using median analyst forecast (SUEAF) in columns 4-6, and the cumulative 3-day abnormal return around the earnings announcement (CAR) in columns 7-9. Control variables are the natural log of lagged quarterly market capitalization (SIZE) and the natural log of the lagged quarterly Book-to-Market ratio (BM). Standard errors are clustered at the industry-year level associated *t*-statistics are reported in parentheses. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| Dep var: | SUE | SUE | SUE | SUEAF | SUEAF | SUEAF | CAR | CAR | CAR |
| IL | 0.12 | 0.04 | 0.03 | -0.01 | 0.01 | 0.02 | -0.14*** | -0.14** | -0.09 |
| | (1.34) | (0.36) | (0.33) | (-1.01) | (0.74) | (1.20) | (-2.89) | (-2.65) | (-1.17) |
| SIZE | | -0.04 | -0.05 | | -0.04*** | -0.04*** | | 0.02 | 0.02 |
| | | (-0.84) | (-1.12) | | (-7.08) | (-6.59) | | (0.59) | (0.41) |
| BM | | -0.68*** | -0.74*** | | 0.04 | 0.03 | | 0.15* | 0.12* |
| | | (-4.84) | (-4.73) | | (1.61) | (1.03) | | (2.03) | (1.84) |
| Obs | 205,477 | 198,987 | 198,987 | 249,541 | 239,512 | 239,512 | 249,058 | 239,031 | 239,031 |
| R-squared | 0.00 | 0.02 | 0.02 | 0.00 | 0.01 | 0.01 | 0.00 | 0.00 | 0.00 |
| Time FE | No | Yes | Yes | No | Yes | Yes | No | Yes | Yes |
| Industry FE | No | No | Yes | No | No | Yes | No | No | Yes |

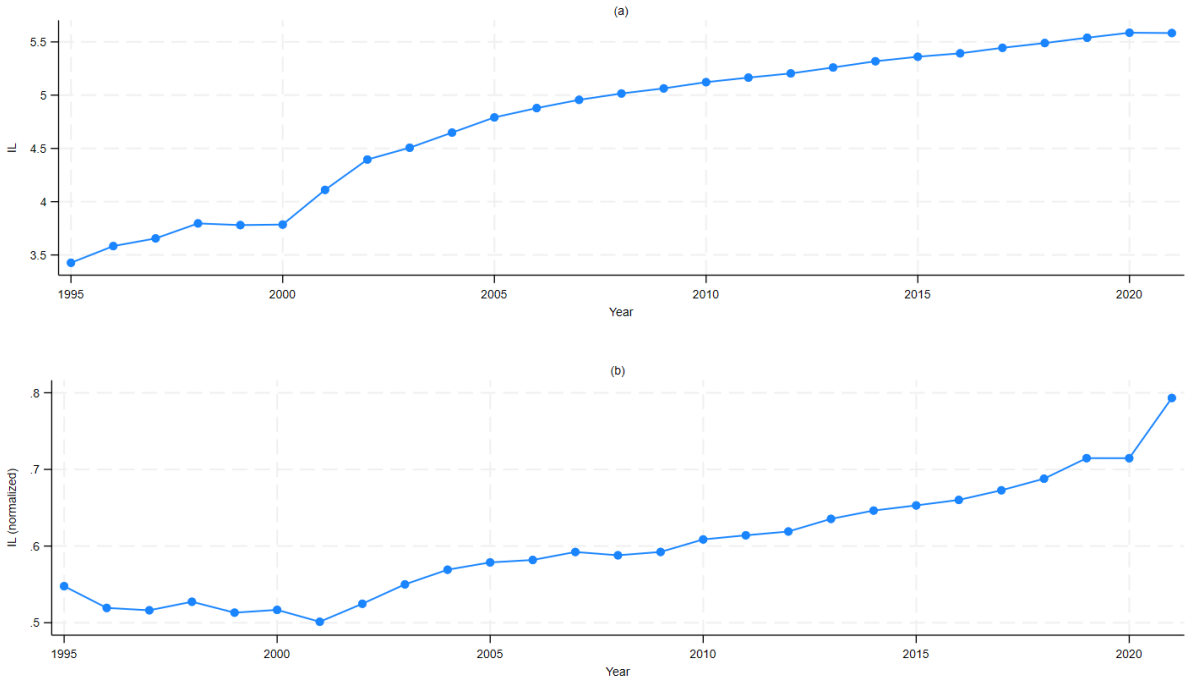


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# Appendix

## Appendix Figure 1. Intangible liabilities over time

This figure plots average intangible liabilities, IL, over time. Subfigure (a) shows IL as measured by the log frequency of words in the IL dictionary that appeared in companies' annual filings. Subfigure (b) shows the frequency of words in the IL dictionary that appeared in companies' annual filings divided by the total words in the filings. The plots show the average of these statistics across firms for each year.






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Appendix Table 1. Words in the IL dictionary

ABROGATE
ABROGATED
ABROGATES
ABROGATING
ABROGATION
ABROGATIONS
ABSOLVE
ABSOLVED
ABSOLVES
ABSOLVING
ACQUIT
ACQUITS
ACQUITTAL
ACQUITTALS
ACQUITTANCE
ACQUITTANCES
ACQUITTED
ACQUITTING
ADJUDGE
ADJUDGED
ADJUDGES
ADJUDGING
ADJUDICATE
ADJUDICATED
ADJUDICATES
ADJUDICATING
ADJUDICATION
ADJUDICATIONS
ADJUDICATIVE
ADJUDICATOR
ADJUDICATORS
ADJUDICATORY
AFFIDAVIT
AFFIDAVITS
AFFIRMANCE
AGGRIEVED
ALLEGATION
ALLEGATIONS
ALLEGE
ALLEGED
ALLEGEDLY
ALLEGES
ALLEGING
ANTICOMPETITIVE
ANTICORRUPTION
ANTITRUST
APPEAL
APPEALABLE
APPEALED
APPEALING
APPEALS
APPELLANT
APPELLANTS
APPELLATE
APPELLEES
APPOINTOR
APPURTENANCE
APPURTENANCES
APPURTENANT
ARBITRABILITY
ARBITRAL
ARBITRATE
ARBITRATED
ARBITRATES
ARBITRATING
ARBITRATION
ARBITRATIONAL
ARBITRATIONS
ARBITRATIVE
ARBITRATOR
ARBITRATORS
ARREARAGE
ARREARAGES
ATTORN
ATTORNEY
ATTORNEYS
ATTORNMENT
ATTORNS
BAIL
BAILED
BAILEE
BAILEES
BAILIFF
BAILIFFS
BAILMENT
BREACH
BREACHED
BREACHES
BREACHING
CERTIORARI
CHATTEL
CHATTELS
CIVIL ACTION
CIVIL PAYMENT
CLAIM
CLAIMABLE
CLAIMANT
CLAIMANTS
CLAIMHOLDER
CLAIMS
CLASS ACTION
CLASS ACTIONS
CLAWBACKS
CODEFENDANT
CODEFENDANTS
COLLUSION
COMPENSATORY
COMPLAINANT
COMPLAINANTS
COMPLAINT
COMPLAINTS
CONDEMNOR
CONFISCATORY
CONSTITUTIONAL
CONSTITUTIONALITY
CONSTITUTIONALLY
CONSTITUTIONS
CONSTRUE
CONSTRUED
CONSTRUES
CONSTRUING
CONTEST
CONTESTABILITY
CONTESTATION
CONTINGENCIES
CONTINGENCY
CONTINGENT
CONTRAVENE
CONTRAVENED
CONTRAVENES
CONTRAVENING
CONTRAVENTION
CONTRAVENTIONS
CONTROVERT
CONTROVERTED
CONTROVERTING
CONVICT
CONVICTED
CONVICTING
CONVICTION
CONVICTIONS
COUNTERCLAIM
COUNTERCLAIMS
COUNTERFEIT
COUNTERFEITER
COUNTERFEITS
COUNTERSUED
COUNTERSUIT
COUNTERSUITS
COURT
COURTROOM
COURTS
CRIME
CRIMES
CRIMINAL
CRIMINALITY
CRIMINALIZE
CRIMINALIZING
CRIMINALLY
CRIMINALS
CROSSCLAIM
CROSSCLAIMS
DAMAGE
DAMAGES
DECEDENT
DECEDENTS
DECEPTIVE
DECLARANT
DECREE
DECREED
DECREEING
DECREES
DEFALCATION
DEFALCATIONS
DEFEASANCE
DEFEASANCES
DEFEASE
DEFEASED
DEFEASEMENT
DEFEASES
DEFEASING
DEFECTIVELY
DEFECTS
DEFENDABLE
DEFENDANT
DEFENDANTS
DEMURRED
DEMURRER
DEMURRERS
DEMURRING
DEMURS
DEPOSE
DEPOSED
DEPOSES
DEPOSING
DEPOSITION
DEPOSITIONAL
DEPOSITIONS
DEROGATE
DEROGATED
DEROGATES
DEROGATING
DEROGATION
DEROGATIONS
DESIST
DETAINER
DISAFFIRM
DISAFFIRMANCE
DISAFFIRMED
DISAFFIRMS
DISPUTE
DISPUTED
DISPUTES
DISTRAINT
DISTRIBUTEE
DISTRIBUTEES
DOCKET
DOCKETED
DOCKETING
DOCKETS
ENCUMBER
ENCUMBERED
ENCUMBERING
ENCUMBERS
ENCUMBRANCE
ENCUMBRANCER
ENCUMBRANCERS
ENCUMBRANCES
ENFORCEABILITY
ENFORCEABLE
ENFORCEABLY
ENVIRONMENTAL
ENVIRONMENTAL
ENVIRONMENTAL
ENVIRONMENTAL
ESCHEAT
ESCHEATED
ESCHEATMENT
EVIDENTIAL
EVIDENTIARY
EXCEEDANCE
EXCEEDANCES
EXCEEDENCES
EXCULPATE
EXCULPATED
EXCULPATES
EXCULPATING
EXCULPATION
EXCULPATIONS
EXCULPATORY
EXECUTOR
EXECUTORS
EXECUTORY
EXECUTRICES
EXECUTRIX
EXTRACONTRACTUAL
EXTRAJUDICIAL
FACIE
FALSE ADVERTISING
FELONIES
FELONIOUS
FELONY
FILED CASE
FILED CASES
FILED SUIT
FINES
FORFEITABILITY
FORFEITABLE
FRAUD
FRAUDS
FRAUDULENT
FUGITIVE
FUGITIVES
HAZARDOUS
HAZARDOUS WASTES
IMPLEADED
INDEMNIFIABLE
INDEMNIFICATION
INDEMNIFICATIONS
INDEMNIFIED
INDEMNIFIES
INDEMNIFY
INDEMNIFYING
INDEMNITEE
INDEMNITEES
INDEMNITIES
INDEMNITOR
INDEMNITORS
INDEMNITY
INDICT
INDICTABLE
INDICTED
INDICTING
INDICTMENT
INDICTMENTS
INFRACTION
INFRACTIONS
INFRINGE
INFRINGED
INFRINGEMENT
INFRINGEMENTS
INFRINGER
INFRINGES
INJUNCTION
INJUNCTIONS
INJUNCTIVE
INJURY
INVESTIGATION
IRREVOCABILITY
IRREVOCABLE
IRREVOCABLY
JUDGMENT
JUDICIAL
JUDICIALLY
JUDICIARIES



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JUDICIARY
JURIES
JURIS
JURISDICTION
JURISDICTIONAL
JURISDICTIONALLY
JURISDICTIONS
JURISPRUDENCE
JURIST
JURISTS
JUROR
JURORS
JURY
JURYMAN
JUSTICE
JUSTICES
LAWSUIT
LAWSUITS
LEGAL
LEGALESE
LEGALITY
LEGALIZATION
LEGALIZATIONS
LEGALIZE
LEGALIZED
LEGALIZES
LEGALIZING
LEGALLY
LEGALS
LEGATEE
LEGATEES
LEGISLATE
LEGISLATED
LEGISLATES
LEGISLATING
LEGISLATION
LEGISLATIONS
LEGISLATIVE
LEGISLATIVELY
LEGISLATOR
LEGISLATORS
LEGISLATURE
LEGISLATURES
LIABILITY
LIABLE
LITIGANT
LITIGANTS
LITIGATE
LITIGATED
LITIGATES
LITIGATING
LITIGATION
LITIGATIONS
LITIGATOR
LITIGATORS
LITIGIOUS
LITIGIOUSNESS
MAJEURE
MANDAMUS
MEDIATE
MEDIATED
MEDIATES
MEDIATING
MEDIATION
MEDIATIONS
MEDIATOR
MEDIATORS
MISAPPROPRIATION
MISDEMEANOR
MISFEASANCE
MISREPRESENTATION
MISREPRESENTATIONS
MISTRIAL
MISTRIALS
MOTIONS
NEGLIGENCE
NONCOMPLIANCE
NONCONTINGENT
NONFEASANCE
NONFIDUCIARY
NONFORFEITABILITY
NONFORFEITABLE
NONFORFEITURE
NONINFRINGEMENT
NONINFRINGING
NONJUDICIAL
NONJUDICIALLY
NONJURISDICTIONAL
NULLIFICATION
NULLIFICATIONS
NULLIFIED
NULLIFIES
NULLIFY
NULLIFYING
NULLITIES
NULLITY
PATENTEE
PENALTIES
PENALTY
PERJURY
PERPETRATE
PERPETRATED
PERPETRATES
PERPETRATING
PERPETRATION
PETITION
PETITIONED
PETITIONER
PETITIONERS
PETITIONING
PETITIONS
PLAINTIFF
PLAINTIFFS
PLEADING
PLEADINGS
PLEADS
POSTJUDGMENT
PREHEARING
PREPETITION
PRETRIAL
PROBATE
PROBATED
PROBATES
PROBATING
PROBATION
PROBATIONAL
PROBATIONARY
PROBATIONER
PROBATIONERS
PROBATIONS
PRODUCT LIABILITIES
PRODUCT LIABILITY
PROFESSIONAL
PROFESSIONAL LIABILITY
PROMULGATE
PROMULGATED
PROMULGATES
PROMULGATING
PROMULGATION
PROMULGATIONS
PROMULGATOR
PROMULGATORS
PROSECUTE
PROSECUTED
PROSECUTES
PROSECUTING
PROSECUTION
PROSECUTIONS
PROSECUTOR
PROSECUTORIAL
PROSECUTORS
REARGUMENT
REBUT
REBUTS
REBUTTABLE
REBUTTABLY
REBUTTAL
REBUTTALS
REBUTTED
REBUTTING
RECALL
RECALLS
REHEAR
REHEARD
REHEARING
REHEARINGS
REMAND
REMANDED
REMANDING
REMANDS
REMEDIATE
REMEDIATED
REMEDIATING
REMEDIATION
REMEDIATIONS
REMEDIED
REQUESTER
REQUESTOR
RESCIND
RESCINDED
RESCINDING
RESCINDS
RESCISSION
RESCISSIONS
RESTITUTION
RESTITUTIONARY
REVOCATION
REVOKE
REVOKED
RULING
RULINGS
SENTENCED
SENTENCING
SETTLE
SETTLEMENT
SETTLEMENTS
SUBDOCKET
SUBLEASEE
SUBLEASEHOLD
SUBLESSORS
SUBPOENA
SUBPOENAED
SUBPOENAS
SUBROGATED
SUBROGATION
SUE
SUED
SUES
SUING
TESTAMENTARY
TESTIFY
TESTIFYING
TESTIMONY
TORT
TORTIOUS
TORTIOUSLY
TORTS
UNAPPEALABLE
UNAPPEALED
UNCONSTITUTIONAL
UNCONSTITUTIONALITY
UNCONSTITUTIONALLY
UNDISCHARGED
UNENCUMBER
UNENCUMBERED
UNENFORCEABILITY
UNENFORCEABLE
UNFAIR
UNLAWFUL
UNLAWFULLY
UNLAWFULNESS
UNREMEDIATED
UNSTAYED
VERDICT
VERDICTS
VIOLATE
VIOLATED
VIOLATING
VIOLATION
VIOLATIVE
WARRANTY DEFECT
WARRANTY DEFECTS
WARRANTY
WARRANTY LIABILITY
WHISTLEBLOWER
WHISTLEBLOWERS
WILFUL
WILLFUL
WILLFULLY
WILLFULNESS
WITNESS
WITNESSES
WRIT
WRITS



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Appendix Table 2. Variable Definitions

| Variable | Definition |
|---|---|
| IL | Our main measure of intangible liabilities. It is the log frequency of IL dictionary words (provided in Appendix Table 1) that appear in each annual report for each firm. |
| UNIQUE | The number of unique words in the IL dictionary (provided in Appendix Table 1) that occur in the 10-K annual report for each firm in each year. |
| PE | Price-earnings ratio calculated as the fiscal-year end stock price per share divided by basic earnings per share. |
| MB | Market-to-book ratio calculated as the market value of equity (stock price at fiscal-year end multiplied by the number of shares outstanding) divided by the book value of equity. |
| SIZE | The logarithm of the market value of equity in millions of dollars calculated as the stock price at fiscal-year end multiplied by the number of shares outstanding. |
| ROA | Return on assets calculated as net income divided by total assets. |
| LEVERAGE | The sum of long-term and short-term debt divided by total assets. |
| TANGIBILITY | Property, plant and equipment divided by total assets. |
| TOTAL WORDS | The logarithm of the total number of words in each annual filing |
| ABNORMAL RET | The yearly stock return in excess of the market return |
| VOLATILITY | Standard deviation of stock returns |
| TURNOVER | Trade volume divided to shares outstanding at the beginning of the year |
| TOBIN Q | Tobin's Q calculated as the sum of market cap of equity and debt divided by the sum of book value of equity and debt. |
| LAWSUIT | An indicator variable equal to one if a firm is subject to a class action lawsuit, and zero otherwise. |
| RRI | An index ranging between 0 and 1, as composed by the RepRisk database on a monthly basis capturing negative media attention about a firm's environmental, social and governance (ESG) realties, practices and issues, based on local, national and international press coverage. RRI in the analyses of this paper are the fiscal year's last month RRI value. |
| BETA | CAPM beta computed using a 5-year rolling window. |
| SIZE | The logarithm of market capitalization, computed by the number of shares outstanding multiplied by the stock price, both from CRSP. |
| BM | The logarithm of book equity from Compustat divided to market cap from CRSP. |
| MOM | Cumulative past 12-month return from CRSP. |
| PROFIT | Operating profitability computed by (revenue – cost of goods sold – SG&A – interest expense) divided to book equity. |
| INV | Asset growth computed by changes in total assets divided to lagged total assets. |
| VOL | The logarithm of past dollar trading volume computed by lagged volume multiplied by price both from CRSP. |
| IDIOVOL | Idiosyncratic volatility in the CAPM model using daily CRSP returns |
| SKEW | Return skewness using daily CRSP data. |
| ILLIQ | Using CRSP daily data, compute the average of daily absolute returns to dollar trading volume per month. ILLIQ is them computed as the average illiquidity during the past 12 months. |
| PRICE | The logarithm of stock price from CRSP. |



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| | |
|---|---|
| YIELD | The ratio of (common dividends plus purchase of common and preferred stocks minus the sale of common and preferred stocks) to lagged market capitalization |
| AdEXP | Advertising expenditures divided to market capitalization |
| R&D | R&D expenditures divided to market capitalization |
| ACCRUALS | [(Changes in current assets) – (changes in cash and short-term investments) – ((changes in current liabilities) – (changes in current debt) – (changes in income tax payable)) – depreciation)]/average total assets |
| REPURCHASE | Purchase of common and preferred stocks |
| NETEQUITY | Sale minus purchase of common and preferred stocks divided by average total assets |
| NCSKEW | The negative conditional return skewness computed following Kim, Li, and Zhang (2011a). |
| CRASH | An indicator variable that is one if a firm experiences at least a crash week in a year and 0 otherwise. A crash week is the week during which firm-specific weekly return ($w_{j,t}$) is 3.2 standard deviations below the average ($w_{j,t}$) over the whole fiscal year. |
| CRASH_EXTR | The negative of the worst deviation of firm specific weekly returns ($w_{j,t}$) from the average $w_{j,t}$ scaled by the standard deviation of $w_{j,t}$ over the entire fiscal year, following (Andreou, Louca, and Petrou 2016). |
| DTURN | De-trended trading volume |
| RET | The average firm-specific weekly return over previous fiscal year |
| SIGMA | standard deviation of $w_{j,t}$ over the fiscal year multiplied by 100 |
| ACCM | Accrual manipulation, measured as the three-year moving sum of absolute discretionary accruals following the modified Jones model (Dechow, Sloan, and Sweeney 1995). |



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Appendix Table 3. Intangible liabilities and future lawsuits: further controls

This table reports results similar to regressions in Table 5 but adds a measure of litigation risk as in Lee et al. (2020).

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Dep variable: | Lawsuit (t+1) | | | | | | | |
| | Panel A. Logit Regressions | | | | | | | |
| IL | 0.66*** | 0.70*** | 0.75*** | 0.95*** | 1.25*** | 1.24*** | 1.26*** | 1.54*** |
| | (0.08) | (0.08) | (0.07) | (0.08) | (0.12) | (0.12) | (0.12) | (0.15) |
| LitRisk | 5.14*** | 5.27*** | 3.33*** | 3.30*** | 5.33*** | 5.32*** | 3.11*** | 3.04*** |
| | (0.37) | (0.34) | (0.28) | (0.27) | (0.30) | (0.30) | (0.36) | (0.37) |
| | Panel B. Linear Probability Model | | | | | | | |
| IL | 1.05*** | 1.15*** | 1.09*** | 1.80*** | 2.25*** | 2.25*** | 2.13*** | 2.84*** |
| | (0.16) | (0.17) | (0.15) | (0.22) | (0.20) | (0.20) | (0.20) | (0.26) |
| LitRisk | 16.14*** | 17.33*** | 11.84*** | 11.76*** | 18.47*** | 18.56*** | 11.78*** | 11.64*** |
| | (1.41) | (1.37) | (1.21) | (1.16) | (1.10) | (1.10) | (1.14) | (1.14) |
| SIZE | | -0.33*** | -0.05 | -0.05 | | -0.52*** | -0.26 | -0.24 |
| | | (0.06) | (0.12) | (0.13) | | (0.17) | (0.22) | (0.22) |
| ABNORMAL RET | | | -1.82*** | -1.84*** | | | -1.66*** | -1.68*** |
| | | | (0.24) | (0.25) | | | (0.22) | (0.22) |
| VOLATILITY | | | 7.81*** | 8.39*** | | | 10.16*** | 10.54*** |
| | | | (2.73) | (2.75) | | | (2.38) | (2.38) |
| TURNOVER | | | 4.48*** | 4.73*** | | | 7.56*** | 7.66*** |
| | | | (0.75) | (0.79) | | | (1.09) | (1.09) |
| LEVERAGE | | | -1.16 | -0.62 | | | -1.41 | -1.10 |
| | | | (0.79) | (0.76) | | | (1.16) | (1.16) |
| ROA | | | -1.24 | -1.50 | | | -4.44*** | -4.68*** |
| | | | (1.15) | (1.10) | | | (1.17) | (1.17) |
| TANGIBILITY | | | 0.69 | 0.47 | | | -0.03 | -0.28 |
| | | | (0.48) | (0.44) | | | (0.68) | (0.67) |
| TOTAL WORDS | | | | -1.90*** | | | | -2.00*** |
| | | | | (0.41) | | | | (0.42) |
| Observations | 27,271 | 27,271 | 27,085 | 27,085 | 27,271 | 27,271 | 27,085 | 27,085 |
| R-squared | 0.06 | 0.06 | 0.07 | 0.07 | 0.04 | 0.04 | 0.05 | 0.06 |
| Industry FE | Yes | Yes | Yes | Yes | Implied | Implied | Implied | Implied |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Firm FE | No | No | No | No | Yes | Yes | Yes | Yes |



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## Appendix Table 4. Intangible liabilities, cost of debt, and audit fees

This table shows the relation between intangible liabilities, IL, and (a) the cost of debt (columns 1-2) and (b) audit fees. Cost of debt is computed using the interest expense divided to total debt. Audit fees are obtained from the Audit Analytics dataset and the logarithm of it is used in regressions. All regressions include year and industry fixed effects. Standard errors are clustered at the industry level and reported in parentheses. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Dep var: | Cost of debt | | Audit Fees | |
| IL | 0.49*** | 0.32*** | 0.21*** | 0.17*** |
| | (0.06) | (0.05) | (0.04) | (0.02) |
| SIZE | -0.78*** | -0.56*** | 0.41*** | 0.43*** |
| | (0.05) | (0.04) | (0.01) | (0.01) |
| MB | | 0.11*** | | -0.04*** |
| | | (0.01) | | (0.00) |
| LEVERAGE | | -9.97*** | | 1.06*** |
| | | (0.69) | | (0.08) |
| ROA | | -3.92*** | | -0.27*** |
| | | (0.27) | | (0.06) |
| TANGIBILITY | | -0.26 | | 0.09 |
| | | (0.20) | | (0.06) |
| TOTAL WORDS | 0.26* | 0.54*** | 0.33*** | 0.28*** |
| | (0.15) | (0.12) | (0.09) | (0.04) |
| Industry FE | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes |
| Observations | 78,228 | 71,798 | 82,525 | 66,546 |
| R-squared | 0.13 | 0.20 | 0.74 | 0.76 |



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Appendix Table 5. Intangible liabilities and abnormal returns: Short versus long legs

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Factor model: | | CAPM | | | FF3 | |
| IL: | Low | High | High - Low | Low | High | High - Low |
| Alpha | -2.83* | 2.06** | 4.89** | -3.45*** | 2.02** | 5.47*** |
| | (-1.68) | (2.12) | (2.57) | (-2.72) | (2.27) | (3.24) |
| MARKET | 0.95*** | 0.89*** | -0.06 | 0.96*** | 0.92*** | -0.04 |
| | (21.01) | (31.96) | (-1.51) | (20.73) | (41.87) | (-0.76) |
| SMB | | | | 0.65 | -2.08*** | -2.73*** |
| | | | | (0.98) | (-6.20) | (-5.23) |
| HML | | | | 4.79*** | 0.67 | -4.12*** |
| | | | | (6.33) | (1.09) | (-4.04) |
| Observations | 306 | 306 | 306 | 306 | 306 | 306 |



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Appendix Table 6. Fama-MacBeth regressions controlling for a litigation risk measure

This table reports results similar to regressions in Table 8 but adds a measure of litigation risk as in Lee et al. (2020), LITRISK. *t*-statistic using Newey standard errors are reported in parentheses. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dep var: | | | Returns (t+1) | | | |
| IL | 0.18*** | 0.19*** | 0.18*** | 0.14** | 0.12* | 0.16*** |
| | (3.10) | (3.34) | (3.05) | (2.36) | (1.89) | (2.68) |
| LITRISK | 0.22** | 0.22** | 0.16* | 0.05 | -0.00 | 0.01 |
| | (2.07) | (2.11) | (1.81) | (0.79) | (-0.08) | (0.23) |
| BETA | 0.03 | -0.03 | -0.00 | -0.10 | -0.11 | -0.14 |
| | (0.18) | (-0.17) | (-0.03) | (-0.85) | (-1.02) | (-1.43) |
| SIZE | -0.38** | -0.38** | -0.29** | -0.41* | -0.33 | -0.50*** |
| | (-2.09) | (-2.15) | (-2.07) | (-1.67) | (-1.25) | (-2.87) |
| BM | 0.25 | 0.30** | 0.32** | 0.21** | 0.13 | 0.16 |
| | (1.36) | (1.99) | (2.59) | (2.05) | (1.21) | (1.17) |
| MOM | | 0.14 | 0.12 | 0.19* | 0.23** | 0.22** |
| | | (1.32) | (1.17) | (1.89) | (2.17) | (2.24) |
| PROFIT | | | -0.08 | -0.01 | 0.08 | 0.12 |
| | | | (-0.56) | (-0.10) | (0.69) | (1.35) |
| INV | | | -0.37*** | -0.29*** | -0.22*** | -0.19*** |
| | | | (-3.84) | (-4.02) | (-3.34) | (-2.62) |
| VOL | | | | 0.73** | 0.64** | 0.78*** |
| | | | | (2.35) | (2.07) | (2.85) |
| IDIOVOL | | | | -0.24* | -0.28** | -0.25** |
| | | | | (-1.94) | (-2.38) | (-2.42) |
| SKEW | | | | 0.02 | 0.02 | 0.02 |
| | | | | (0.48) | (0.40) | (0.48) |
| ILLIQ | | | | 0.10 | 0.08 | 0.21** |
| | | | | (0.56) | (0.41) | (1.98) |
| PRICE | | | | -0.85*** | -0.71*** | -0.73*** |
| | | | | (-3.51) | (-3.10) | (-3.05) |
| YIELD | | | | -0.01 | 0.01 | 0.12 |
| | | | | (-0.18) | (0.09) | (1.20) |
| AdEXP | | | | | -0.00 | -0.07 |
| | | | | | (-0.06) | (-0.75) |
| R&D | | | | | 0.51*** | 0.46*** |
| | | | | | (4.29) | (3.46) |
| LEVERAGE | | | | | | 0.08 |
| | | | | | | (0.46) |
| ACCRUALS | | | | | | -0.11** |
| | | | | | | (-2.34) |
| REPURCHASE | | | | | | 0.02 |
| | | | | | | (0.57) |
| NETEQUITY | | | | | | 0.11 |
| | | | | | | (1.53) |
| TOTAL WORDS | -0.05 | -0.04 | -0.01 | -0.09 | -0.11* | -0.13** |
| | (-0.72) | (-0.57) | (-0.08) | (-1.44) | (-1.85) | (-2.12) |
| Observations | 321,529 | 321,362 | 320,641 | 320,567 | 320,567 | 279,800 |
| R-squared | 0.05 | 0.06 | 0.06 | 0.08 | 0.09 | 0.09 |



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Appendix Table 7. Fama-MacBeth regressions controlling for two litigation risk measures

This table reports results similar to regressions in Table 8 but adds two measures of litigation risk. LITRISK is the measure introduced by Lee et al. (2020), whereas PLFVIEWS is from Kartapanis and Yust (2024). *t*-statistics using Newey standard errors are reported in parentheses. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dep var: | | | Returns (t+1) | | | |
| IL | 0.19 | 0.20* | 0.20* | 0.24*** | 0.23*** | 0.21*** |
| | (1.65) | (1.77) | (1.95) | (2.86) | (3.16) | (2.69) |
| LITRISK | -0.00 | -0.00 | -0.04 | -0.14* | -0.19** | -0.14* |
| | (-0.03) | (-0.00) | (-0.41) | (-1.90) | (-2.49) | (-1.90) |
| PLFVIEWS | 0.08 | 0.08 | 0.06 | -0.05 | -0.13 | 0.03 |
| | (1.37) | (1.31) | (1.19) | (-0.75) | (-1.02) | (0.69) |
| BETA | 0.06 | 0.01 | -0.03 | -0.09 | -0.06 | -0.08 |
| | (0.32) | (0.09) | (-0.15) | (-0.70) | (-0.52) | (-0.73) |
| SIZE | -0.08 | -0.08 | -0.01 | 0.12 | 0.46 | -0.23 |
| | (-0.63) | (-0.64) | (-0.08) | (0.23) | (0.63) | (-0.90) |
| BM | 0.30 | 0.26 | 0.28* | 0.20* | 0.19* | 0.17 |
| | (1.45) | (1.57) | (1.84) | (1.73) | (1.67) | (1.05) |
| MOM | | -0.11 | -0.11 | -0.08 | -0.07 | -0.01 |
| | | (-0.84) | (-0.86) | (-0.69) | (-0.51) | (-0.09) |
| PROFIT | | | -0.11 | -0.08 | -0.10 | 0.09 |
| | | | (-0.61) | (-0.45) | (-0.43) | (0.82) |
| INV | | | -0.29** | -0.20*** | -0.11 | -0.11 |
| | | | (-2.42) | (-3.04) | (-1.65) | (-1.46) |
| VOL | | | | 0.35 | 0.04 | 0.60* |
| | | | | (0.69) | (0.07) | (1.67) |
| IDIOVOL | | | | 0.02 | 0.01 | -0.06 |
| | | | | (0.14) | (0.09) | (-0.46) |
| SKEW | | | | -0.01 | -0.03 | 0.01 |
| | | | | (-0.19) | (-0.44) | (0.22) |
| ILLIQ | | | | -0.37 | -0.59 | -0.03 |
| | | | | (-1.06) | (-1.09) | (-0.41) |
| PRICE | | | | -0.61** | -0.52* | -0.42 |
| | | | | (-2.17) | (-1.82) | (-1.58) |
| YIELD | | | | -0.02 | -0.06 | 0.01 |
| | | | | (-0.20) | (-0.48) | (0.06) |
| AdEXP | | | | | 0.14 | 0.02 |
| | | | | | (1.44) | (0.11) |
| R&D | | | | | 0.28** | 0.26* |
| | | | | | (2.14) | (1.81) |
| LEVERAGE | | | | | | -0.15 |
| | | | | | | (-0.42) |
| ACCRUALS | | | | | | -0.11* |
| | | | | | | (-1.84) |
| REPURCHASE | | | | | | 0.03 |
| | | | | | | (0.66) |
| NETEQUITY | | | | | | 0.07 |
| | | | | | | (0.58) |
| TOTAL WORDS | -0.04 | -0.04 | -0.03 | -0.16** | -0.17** | -0.16** |
| | (-0.34) | (-0.33) | (-0.22) | (-2.00) | (-2.27) | (-2.17) |
| Observations | 126,688 | 126,669 | 126,527 | 126,516 | 126,516 | 114,944 |
| R-squared | 0.05 | 0.05 | 0.06 | 0.08 | 0.09 | 0.10 |



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## Appendix Table 8. Fama-MacBeth regressions controlling for intangible capital

This table reports results similar to regressions in Table 8 but adds (different measures) of intangible capital as additional variables on the right-hand side of regressions. Four different measures are used: ORGCAP EP is the organization capital as in Eisfeldt and Papanikolaou (2013), whereas KNWLEDGE CAP PT, ORGCAP PT and INTCAP PT are knowledge capital, organization capital, and intangible capital of Peters and Taylor (2017), respectively. For each measure, there are two specifications corresponding to the first model (main controls) and the last model (full controls) of Table 8. *t*-statistics using Newey standard errors are reported in parentheses. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Dep var: | Return (t+1) | | | | | | | |
| IL | 0.16*** | 0.17*** | 0.15** | 0.16*** | 0.16** | 0.16*** | 0.14** | 0.16*** |
| | (3.01) | (4.28) | (2.38) | (4.20) | (2.34) | (4.12) | (2.30) | (4.22) |
| ORGCAP EP | 0.28*** | 0.10* | | | | | | |
| | (2.79) | (1.87) | | | | | | |
| KNOWLEDGE CAP PT | | | 0.26* | 0.11 | | | | |
| | | | (1.86) | (1.06) | | | | |
| ORGCAP PT | | | | | 0.14* | 0.01 | | |
| | | | | | (1.91) | (0.29) | | |
| INTCAP PT | | | | | | | 0.26** | 0.04 |
| | | | | | | | (2.16) | (0.57) |
| Controls | Main | Full | Main | Full | Main | Full | Main | Full |



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Appendix Table 9. Time series portfolio return of intangible liabilities: Analyzing different sections

This table shows the abnormal return of long-short portfolios sorted based on intangible liabilities, IL. There are 6 columns representing 6 different factor models. The abnormal returns are reported for intangible liabilities, IL, constructed based on two different sections of the annual filing: (a) Risk Factors section abnormal returns are shown in the first row, and (b) the company filing excluding Risk Factors section's shown in the second row of the table. Excess returns associated with different sections are estimated separately, i.e., each number in the table comes from a different regression depending on the factor model choice and the section based on which the IL is constructed. *t*-statistics using Newey standard errors are reported in parentheses. ***, **, * denote significance at 1%, 5%, and 10% respectively.

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| | CAPM | FF3 | FF6 | q5 | DHS4 | SY4 |
| α (%) – Risk Factors | 5.12** | 5.46*** | 5.42*** | 4.70** | 7.00*** | 8.52*** |
| | (2.10) | (2.88) | (2.90) | (2.42) | (3.11) | (3.11) |
| α (%) – Ex Risk Factors | 4.53*** | 4.86*** | 3.41*** | 3.36** | 4.58*** | 4.17*** |
| | (3.04) | (3.24) | (2.70) | (2.43) | (3.81) | (2.68) |



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Appendix Table 10. Most important words in the IL dictionary

| | | | |
|---|---|---|---|
| INFRINGEMENT | MAJEURE | DEROGATE | SUBLESSORS |
| NULLIFIES | ENCUMBERING | LITIGATION | EXCULPATION |
| PRODUCT LIABILITY | EVIDENTIAL | DISAFFIRMANCE | DISPUTE |
| PROMULGATED | PROSECUTE | ARBITRATOR | REBUTTABLE |
| RESCISSION | PATENTEE | WITNESSES | APPEALS |
| RESCINDS | PROSECUTES | REBUTS | PLEADING |
| JURISDICTIONS | UNENFORCEABILITY | INFRACTIONS | ALLEGE |
| FRAUD | ENVIRONMENTAL MATTERS | COLLUSION | INDEMNITEES |
| MEDIATED | AFFIDAVITS | INDEMNIFIES | CONSTITUTIONS |
| PROSECUTION | VIOLATION | CONTROVERT | LEGISLATURES |
| ENFORCEABILITY | PROMULGATORS | ATTORNEY | CONTRAVENTION |
| LEGAL | AFFIRMANCE | REBUTTABLY | DEFALCATION |
| BAILIFF | MEDIATES | ADJUDGING | SETTLE |
| FELONIES | JUDICIARY | JUDICIALLY | DEMURRERS |
| DEROGATIONS | LEGALLY | REMEDIATING | CRIMES |
| RESCIND | APPEALING | DEPOSED | INJURY |
| CONTINGENCY | CODEFENDANT | INFRINGES | FALSE ADVERTISING |
| UNENCUMBER | ENFORCEABLE | ACQUIT | VIOLATIVE |
| PROSECUTED | VERDICT | CONSTRUES | NULLIFY |
| ATTORNMENT | HAZARDOUS SUBSTANCES | UNENCUMBERED | APPURTENANCES |
| NONFORFEITABLE | CLAIMANTS | DEMURRER | EXECUTORS |
| DEPOSES | BAILED | EXCEEDANCES | PROSECUTING |
| CONDEMNOR | CHATTEL | WARRANTYLIABILITIES | DEMURRED |
| FORFEITABLE | RESTITUTION | RESTITUTIONARY | CONTESTABILITY |
| ENCUMBRANCER | ARBITRATING | SUBROGATION | ABSOLVE |
| REMEDIATE | ADJUDICATED | DISAFFIRM | DEROGATION |
| IRREVOCABLY | CONSTITUTIONALITY | REBUTTALS | UNREMEDIATED |
| CONFISCATORY | UNCONSTITUTIONAL | BREACHED | INDICTING |
| COURTS | COUNTERCLAIM | CONTRAVENES | INDEMNIFICATIONS |
| LIABILITY INSURANCE | LEGALIZE | INFRINGEMENTS | ANTITRUST |
| ABSOLVED | TESTIFY | DISAFFIRMS | MOTIONS |
| JURIST | LEGISLATIONS | ACQUITTING | ARBITRATIONS |
| DESIST | UNAPPEALED | NULLIFIED | LEGALIZING |
| REMEDIATED | CONTINGENT | CONTRAVENED | FORFEITABILITY |
| PROMULGATE | FILEDCASE | REQUESTER | PROBATION |
| SETTLEMENTS | FELONIOUS | INDEMNIFYING | PROMULGATION |
| INFRINGER | SETTLEMENT | REBUTTING | REMEDIED |
| | | | JUDICIAL |



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